AI System for Sinus CT Analysis

Graduation Project 2 prepared to obtain the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science

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SUPERVISOR CERTIFICATION

I certify that the preparation of this project, entitled
"Project Title,"
prepared by
"Students' Names,"
was carried out under my supervision at the Faculty of Artificial Intelligence Engineering in partial fulfillment of the requirements for the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science.

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كلمة شكر وتقدير

إهداء

الملخص

يعالج هذا المشروع مشكلة تحليل صور التصوير الطبقي المحوري للجيوب الأنفية باستخدام تقنيات التعلم العميق. تكمن أهمية المسألة في أن صور الجيوب الأنفية تحتوي على بنى تشريحية متداخلة، وأن تقييمها يدوياً قد يكون مرهقاً ويتطلب خبرة اختصاصية. كما تظهر تحديات إضافية بسبب محدودية البيانات، وعدم توازن الحالات المرضية، وترافق أكثر من شذوذ تشريحي أو مرضي في الحالة الواحدة.

لتحديد مناطق مهمة مثل SwinUNETR يقترح المشروع خط معالجة يعتمد على التقسيم التشريحي أولاً باستخدام نموذج الجيب الفكي الأيمن والأيسر والتجويف الأنفي الأيمن والأيسر والبلعوم الأنفي. بعد ذلك يتم استخراج مناطق اهتمام موجهة ConvNeXt-Tiny تشريحياً، ثم تدريب نموذج تصنيف متعدد التسميات للكشف عن خمس علامات مرضية مختارة. تم اعتماد Swin-T. كنموذج التصنيف النهائي بعد مقارنته مع

مقدار IoU النهائي 0.8743 ومتوسط Dice أظهرت نتائج التقسيم قدرة جيدة على تحديد البنى التشريحية، حيث بلغ متوسط F1 وفق متوسط Swin-T أداء أفضل من ConvNeXt-Tiny على مجموعة الاختبار. أما في التصنيف، فقد حقق 0.8102 على الطيات ونتائج التنبؤ خارج الطية. ومع ذلك، بقيت بعض التسميات، خاصة شذوذ الجيب الفكي، أقل استقراراً بسبب انخفاض عدد العينات الإيجابية وترافقها المتكرر مع شذوذات أخرى.

كلمات مفتاحية: التصوير الطبقي المحوري، الجيوب الأنفية، التعلم العميق، التقسيم الطبي، التصنيف متعدد التسميات.

Abstract

This project addresses the problem of artificial-intelligence-assisted analysis of sinus computed tomography (CT) scans. Sinus CT interpretation is clinically important because it supports the assessment of sinonasal anatomy and pathology, but automatic analysis is challenging due to the three-dimensional nature of CT volumes, anatomical variability, limited annotated data, and strong class imbalance among abnormal findings.

The proposed solution follows a segmentation-assisted pipeline. First, a SwinUNETR model is trained to segment key anatomical structures in NasalSeg CT volumes. Second, anatomy-guided regions of interest (ROIs) are extracted from the CT scan and segmentation masks. Third, a multi-label classifier predicts five final abnormality labels: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy. ConvNeXt-Tiny is selected as the final classifier after comparison with Swin-T.

The segmentation model achieved a mean Dice score of 0.8743 and mean IoU of 0.8102 on the test set. For classification, ConvNeXt-Tiny achieved a fold-level macro F1 score of 0.7849 and an out-of-fold macro F1 score of 0.6928, outperforming Swin-T in the final comparison. The results show that the proposed pipeline is promising, while also revealing important limitations caused by small positive sample counts, multi-label co-occurrence, and indirect anatomical targets such as turbinate abnormalities.

Keywords: Sinus CT, NasalSeg, SwinUNETR, ConvNeXt-Tiny, multi-label classification.

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Glossary of Terms and Symbols

Table 0.1: Glossary of terms and abbreviations

Term	English Equivalent / Description	Symbol / Abbreviation
Computed Tomography	Cross-sectional X-ray imaging modality used to visualize internal anatomy.	CT
Region of Interest	A selected image region used as model input for a specific label.	ROI
Dice Similarity Coefficient	Overlap metric used to evaluate segmentation masks.	Dice
Intersection over Union	Overlap metric between predicted and reference masks.	IoU
Convolutional Neural Network	Deep learning architecture based on convolutional operations.	CNN
Vision Transformer	Neural architecture that applies self-attention to image patches.	ViT
SwinUNETR	Transformer-based 3D U-shaped segmentation architecture.	SwinUNETR
ConvNeXt	Modern convolutional architecture inspired by transformer-era design choices.	ConvNeXt
Out-of-Fold Prediction	Prediction produced for each sample by a model that did not train on it.	OOF
Focal Loss	Classification loss that reduces the contribution of easy examples.	Focal Loss
Average Precision	Area under the precision-recall curve, useful for imbalanced classification.	AP
Exact Match Accuracy	Multi-label metric requiring all labels of a case to be predicted correctly.	Subset Accuracy

Chapter One: Introduction

1.1 Introduction

Computed tomography (CT) is one of the most important imaging modalities for evaluating the paranasal sinuses and nasal cavity. It provides detailed cross-sectional views of bony and soft-tissue structures, making it useful for diagnosing sinus disease, studying anatomical variation, and planning endoscopic sinus surgery. However, CT volumes are three-dimensional and usually contain many slices, which makes manual review time-consuming and dependent on clinical experience.

Artificial intelligence, especially deep learning, has become an important research direction in medical image analysis. Deep learning models can learn visual patterns directly from imaging data and can be trained to perform segmentation, classification, detection, and quantification tasks. In sinus CT analysis, segmentation can localize anatomical structures, while classification can support the detection of abnormal findings. Combining these two tasks may help produce a system that is more anatomically meaningful than a purely image-level classifier.

This project focuses on developing an AI system for sinus CT analysis using a segmentation-assisted classification pipeline. The system first localizes important sinonasal structures using a SwinUNETR segmentation model, then extracts anatomy-guided regions of interest, and finally classifies five selected abnormality labels using modern image-classification backbones. The final classification model selected in this work is ConvNeXt-Tiny because it achieved the strongest performance among the final evaluated classifiers.

1.2 Scientific Problem of the Research Project

The scientific problem addressed in this project is the automatic analysis of sinus CT scans for selected anatomical and pathological abnormalities. The problem is not a simple image classification task because the abnormalities are region-dependent, multi-label, and often correlated. A patient may have septal deviation, turbinate hypertrophy, and maxillary sinus abnormality at the same time. A single-label classifier or a simple dominant-label confusion matrix cannot fully represent this situation.

The problem also includes data-related challenges. The dataset contains a limited number of CT cases, and the positive counts differ strongly across labels. For example, the final labels include 71 positive cases for nasal septum abnormality, 60 positive cases for each inferior turbinate hypertrophy label, but only 22 and 23 positives for the right and left maxillary abnormality labels. This imbalance makes maxillary abnormalities less stable during training and evaluation.

Another difficulty is the mismatch between available segmentation masks and some clinical targets. NasalSeg provides masks for the right and left maxillary sinuses, right and left nasal cavities, and nasopharynx. It does not directly segment the nasal septum or turbinates. Therefore, the final model must infer septal and turbinate-related abnormalities using contextual neighboring regions rather than direct target masks.

1.3 Research Objectives

- Develop an AI-assisted pipeline for sinus CT analysis based on segmentation-guided region extraction.
- Train a 3D SwinUNETR model to segment key nasal and paranasal anatomical structures.
- Convert clinical annotation-sheet findings into binary abnormality labels suitable for multi-label learning.
- Design stable anatomical ROIs that connect segmentation outputs to clinically meaningful classification targets.
- Compare ConvNeXt-Tiny and Swin-T as final classification backbones under a 5-fold evaluation setup.
- Select the best-performing final classifier and analyze its strengths, weaknesses, and failure cases.

1.4 Research Contribution

The main contribution of this project is the development of a practical segmentation-assisted multi-label classification workflow for sinus CT analysis. Instead of classifying the full CT volume blindly, the model uses anatomical localization to extract targeted ROIs. This helps connect each classification output to a relevant anatomical area and makes the pipeline easier to interpret.

A second contribution is the final ROI design. Because NasalSeg does not directly provide septum or turbinate masks, the project explores contextual regions based on the available nasal cavity masks. The final septum ROI is defined using both nasal cavities with padding, representing the idea that the septum lies between the right and left nasal cavity regions. The inferior turbinate labels are modeled using the ipsilateral nasal cavity ROIs.

A third contribution is the comparative evaluation of ConvNeXt-Tiny and Swin-T for this specific multi-label sinus CT task. ConvNeXt-Tiny is selected as the final model because it achieves better fold-level and out-of-fold performance than Swin-T. The report also discusses why some labels, especially maxillary abnormalities, remain unstable due to low positive counts and label co-occurrence.

1.5 Target Beneficiaries

- Radiologists and ENT physicians who interpret sinus CT scans.
- Hospitals and imaging centers interested in AI-assisted triage or decision support.

- Researchers working on medical image segmentation and classification.
- Students and engineers studying applied deep learning in medical imaging.
- Developers who may extend the system into a clinical review or educational tool.

1.6 Report Organization

Chapter Two presents the theoretical background needed to understand the project, including CT imaging, sinus anatomy, segmentation, classification, CNNs, transformers, and evaluation metrics. Chapter Three reviews related work and identifies the research gap. Chapter Four explains the research methodology, dataset, preprocessing, segmentation model, ROI extraction, and final classification models. Chapter Five presents the experimental results and discusses the segmentation and classification outcomes. Chapter Six concludes the report and proposes future work.

Chapter Two: Theoretical Background

2.1 Computed Tomography and Sinus Imaging

Computed tomography constructs cross-sectional images from X-ray measurements acquired around the patient. In sinus imaging, CT is especially valuable because it shows the bony boundaries of the paranasal sinuses and nasal cavity. The intensity values in CT images are commonly represented in Hounsfield Units (HU), where air has very low values and bone has high values. This contrast is useful for distinguishing air-filled cavities from surrounding structures.

Sinus CT volumes are usually analyzed in axial, coronal, and sagittal planes. A model trained on CT data must therefore learn three-dimensional spatial relationships, or approximate them through multi-view two-dimensional representations. In this project, segmentation is performed as a 3D task, while classification uses 2.5D ROI images derived from each anatomical crop.

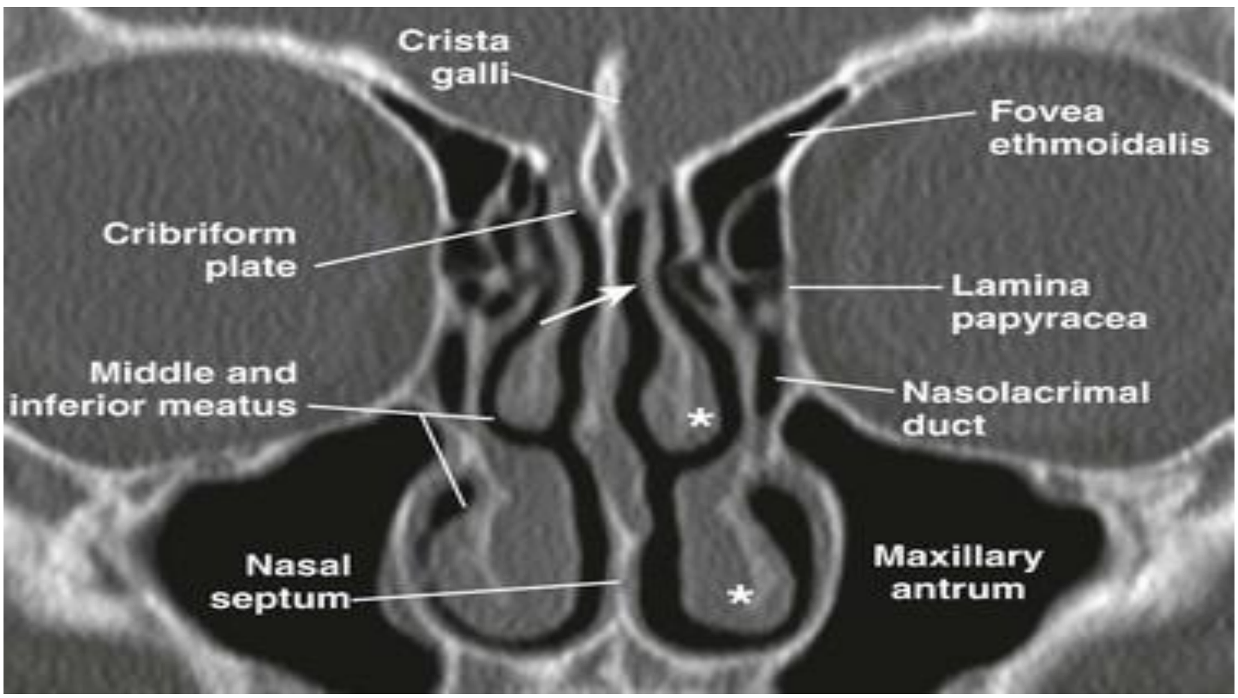

Figure 2.1: Main sinonasal anatomical regions in CT imaging

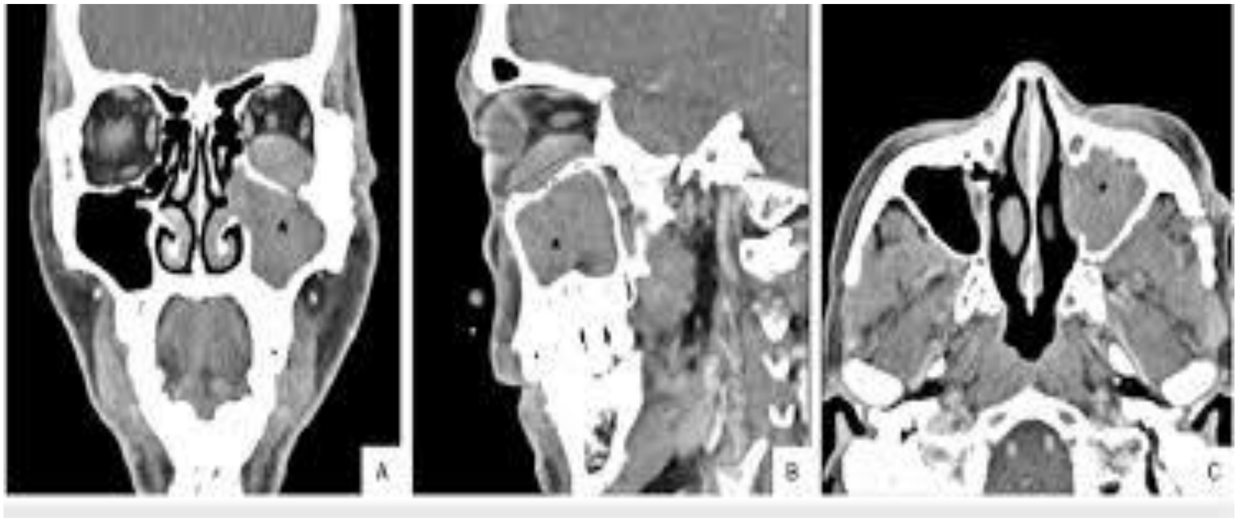

Figure 2.2: Example CT views of the sinonasal area, coronal, sagittal, axial

2.2 Sinonasal Anatomy Relevant to the Project

The paranasal sinuses are air-filled cavities surrounding the nasal cavity. The maxillary sinuses are located on both sides of the nasal cavity and are among the largest paranasal sinuses. The nasal cavity is divided into right and left sides by the nasal septum. The turbinates are curved bony structures covered by mucosa that project into the nasal cavity and affect airflow.

The final classification labels selected in this project are connected to these anatomical regions. Right and left maxillary abnormalities are associated with the corresponding maxillary sinus ROIs. Nasal septum abnormality is associated with the region between the two nasal cavities. Inferior turbinate hypertrophy is associated with the right or left nasal cavity because direct turbinate segmentation masks are not available in the dataset.

2.3 Medical Image Segmentation

Medical image segmentation is the task of assigning a class label to each pixel or voxel. In 3D CT segmentation, each voxel in the volume is assigned to an anatomical class such as background, right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, or nasopharynx. Segmentation is useful because it localizes structures and enables measurement, visualization, and region-specific analysis.

In this project, segmentation is not the final clinical output. Instead, it is an intermediate stage used to guide ROI extraction. This design allows the classification model to focus on relevant anatomical regions rather than the entire CT volume.

$$Dice = 2\,|P \cap G| \,/\, (|P| + |G|)$$

Equation [2.1]: Dice Similarity Coefficient

$$IoU = |P \cap G| \,/\, |P \cup G|$$

Equation [2.2]: Intersection over Union

2.4 Deep Learning for CT Analysis

Deep learning models learn hierarchical representations from data. Early layers usually capture simple intensity and edge patterns, while deeper layers capture more complex anatomical or pathological features. For CT analysis, the model must be robust to anatomical variation, acquisition differences, and class imbalance.

There are two broad approaches to CT learning. A 3D model processes the full volume or volume crops and can preserve volumetric context, but it requires more memory and more data. A 2D or 2.5D model processes slices or multi-plane views, reducing memory requirements and enabling the use of pretrained natural-image backbones. This project uses 3D modeling for segmentation and 2.5D transfer learning for classification.

2.5 Convolutional Neural Networks and ConvNeXt

Convolutional neural networks are well suited for medical images because they exploit local spatial structure. ConvNeXt is a modern convolutional architecture that updates classical CNN design choices while preserving the simplicity and locality of convolutional operations. In this project, ConvNeXt-Tiny is used as the final classification backbone because it showed the best balance between performance and stability on the selected multi-label task.

The selection of ConvNeXt-Tiny is supported by two ideas. First, ConvNeXt preserves convolutional inductive bias, which is often helpful in limited-data medical imaging. Second, its design includes transformer-era improvements while avoiding some of the data demands of pure transformer backbones. This makes it a reasonable candidate for ROI-based 2.5D classification, where the input already focuses on a clinically meaningful region.

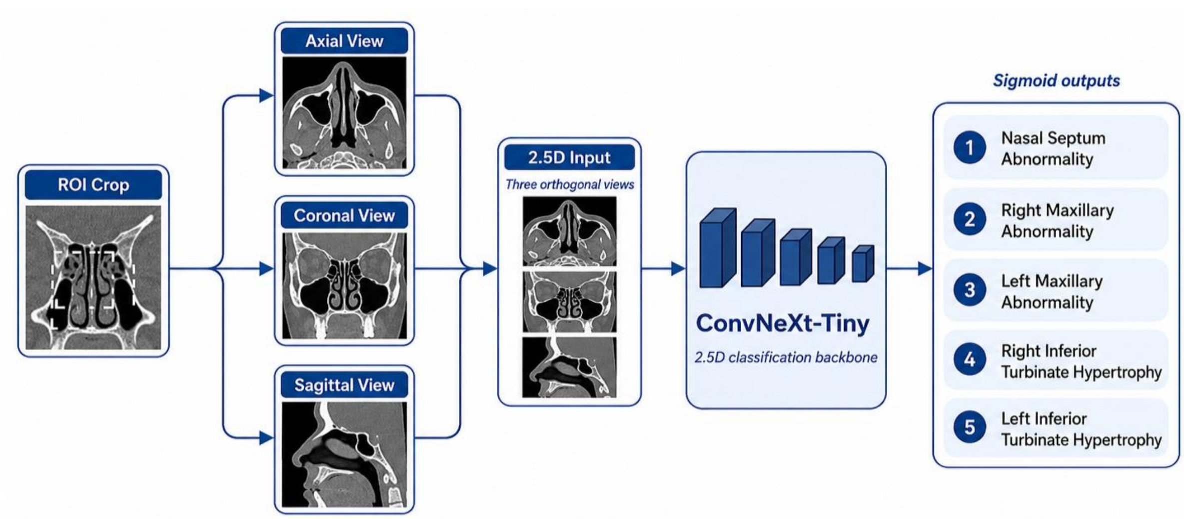

Figure 2.3: Simplified ConvNeXt-Tiny classification concept

2.6 Vision Transformers, Swin Transformer, and SwinUNETR

Vision transformers process images through patch representations and self-attention. Standard transformers can be computationally expensive for high-resolution medical images. Swin Transformer addresses this issue by using hierarchical shifted windows, which makes attention more efficient while still capturing contextual information. SwinUNETR combines a Swin Transformer encoder with a U-shaped segmentation decoder, making it suitable for 3D medical segmentation tasks.

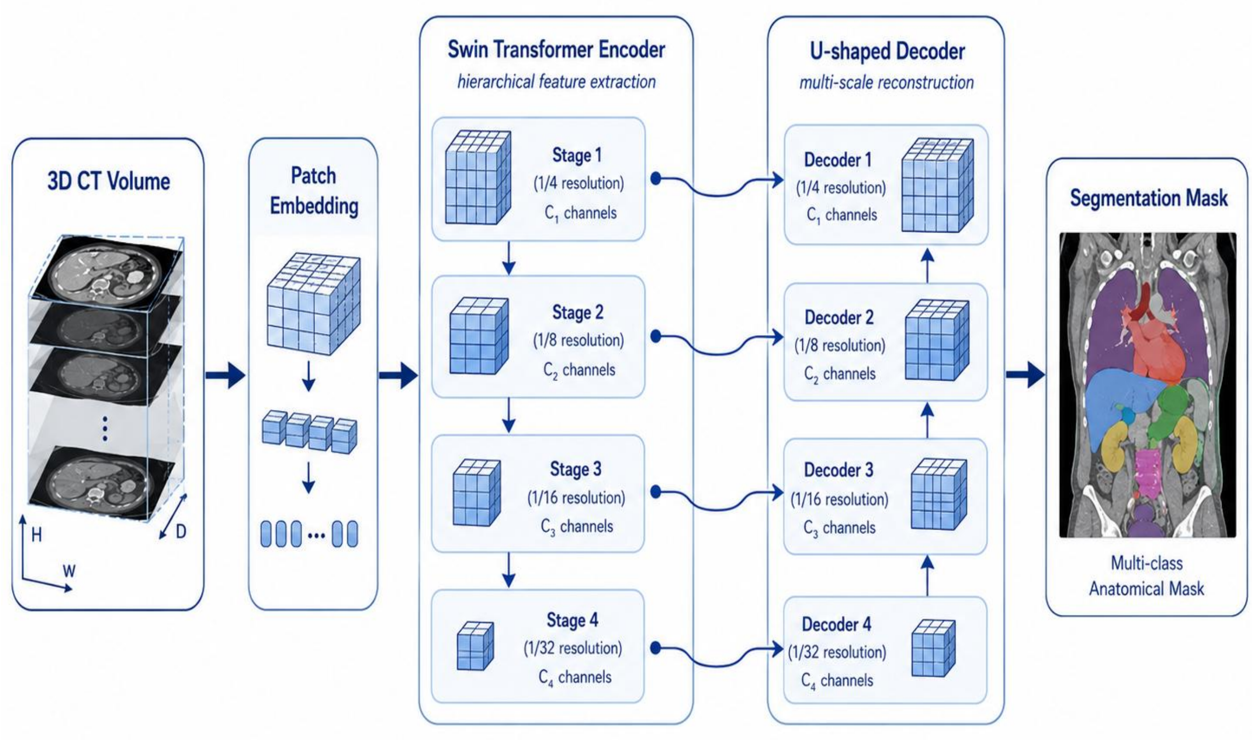

Figure 2.4: Simplified SwinUNETR architecture overview

2.7 Multi-Label Classification

In multi-label classification, each case can belong to more than one class. This is different from multi-class classification, where each sample belongs to exactly one class. The sinus CT task in this project is multi-label because one case may contain multiple simultaneous abnormalities. For this reason, each final label is predicted with an independent sigmoid output rather than a single softmax output.

$$\hat{y}\_i = 1 \text{ if } sigmoid(z\_i) \geq \tau\_i, \text{ otherwise } 0$$

Equation [2.3]: Binary classification decision from sigmoid probability

2.8 Class Imbalance and Focal Loss

Class imbalance occurs when positive and negative examples are unevenly distributed. In the final dataset, maxillary abnormality labels have much fewer positive samples than septum and inferior turbinate labels. This can cause the model to overfit rare labels or predict them inconsistently. Focal loss helps by reducing the effect of easy examples and giving more attention to difficult cases. The final experiments use focal loss with gamma equal to 1 and positive class weighting.

$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

Equation [2.4]: Binary focal loss concept

2.9 Evaluation Metrics

2.9.1 Segmentation Metrics

The segmentation model was evaluated with spatial overlap metrics and runtime measures. Dice and IoU measure agreement between predicted and reference masks. Precision measures the proportion of predicted foreground voxels that are correct, while recall or sensitivity measures the proportion of reference voxels captured by the model. Inference time and throughput describe the practical speed of the trained model during test-time prediction.

$$Precision = TP / (TP + FP)$$

Equation [2.5]: Precision

$$Recall = TP / (TP + FN)$$

Equation [2.6]: Recall / Sensitivity

$$Specificity = TN / (TN + FP)$$

Equation [2.7]: Specificity

2.9.2 Classification Metrics

The classification task was evaluated using complementary metrics because the task is imbalanced and multi-label. Accuracy measures label-wise correctness. Precision and recall measure false-positive and false-negative behavior. Specificity measures the ability to reject negative cases. Balanced accuracy averages sensitivity and specificity, making it more informative when positive and negative cases are imbalanced.

F1-score is the harmonic mean of precision and recall. Macro F1 averages label-wise F1 scores, giving each abnormality equal importance regardless of its frequency. AUC measures ranking quality across thresholds, while average precision summarizes the precision-recall curve and is useful for imbalanced labels. Overall label accuracy measures correctness over all labels, while exact match accuracy requires every label of a case to be predicted correctly.

The report also uses out-of-fold evaluation. In a 5-fold setting, each case is predicted by a model that did not train on it. This produces a more reliable estimate of generalization than evaluating only on the training folds. Threshold tuning is performed on the OOF predictions to select label-specific probability cutoffs. The 5×5 dominant-label confusion matrix is used only as a visualization because it compresses a multi-label prediction into one dominant true label and one dominant predicted label.

$$F1 = 2 \times Precision \times Recall / (Precision + Recall)$$

Equation [2.8]: F1-score

$$Balanced\ Accuracy = (Sensitivity + Specificity) / 2$$

Equation [2.9]: Balanced accuracy

$$Exact\ Match = number\ of\ cases\ with\ all\ labels\ correct / total\ number\ of\ cases$$

Equation [2.10]: Exact match accuracy

2.10 Chapter Summary

This chapter introduced the theoretical foundations of the project, including sinus CT, sinonasal anatomy, segmentation, multi-label classification, CNNs, transformers, class imbalance, and evaluation metrics. These concepts support the methodology described in Chapter Four.

Chapter Three: Literature Review

3.1 Introduction

This chapter reviews previous studies related to sinus CT segmentation, medical image classification, ConvNeXt, Swin Transformer, and 2.5D transfer learning. The purpose is not only to list related papers, but also to justify the methodological choices of this project. In particular, the literature review explains why a segmentation-guided pipeline was selected and why ConvNeXt-Tiny was a reasonable final classification candidate rather than an arbitrary choice.

The reviewed literature shows that sinus CT research has often focused on anatomical segmentation, while classification studies are usually narrower, such as maxillary anomaly classification or general medical-image classification. Therefore, the research gap is a combined pipeline that uses segmentation as an anatomical guide and then performs multi-label abnormality prediction.

3.2 Previous Studies in Sinus CT Segmentation and Analysis

NasalSeg is the most directly related dataset for this project because it provides 130 CT scans with voxel-level annotations of five sinonasal structures. Its main value is that it makes anatomical segmentation research possible on a public dataset. However, it does not directly annotate the nasal septum or turbinates, which creates an important limitation for abnormality classification tasks that require these structures.

Recent paranasal sinus segmentation studies also demonstrate the value of 3D deep learning for anatomical localization. Whangbo et al. compared 3D U-Net variants for paranasal sinus segmentation in sinusitis CT, while Song et al. compared CNN, vision-transformer, and hybrid networks for paranasal sinus segmentation. These studies support the use of deep segmentation models but still focus mainly on anatomical masks rather than downstream multi-label abnormality prediction.

Other studies focus on maxillary sinus segmentation or anomaly classification. Ozturk et al. used U-Net for maxillary sinus segmentation on CBCT, while Bhattacharya et al. explored self-supervised learning for classifying paranasal anomalies in the maxillary sinus. These studies are relevant because maxillary disease is one of the labels in this project, but they do not solve the broader five-label sinus CT classification task.

3.3 Classification Backbones and Transfer Learning Studies

The classification phase of this project used modern transfer-learning backbones. ConvNeXt is especially relevant because it revisits convolutional design using transformer-era improvements while retaining the spatial inductive biases of CNNs. This is important for limited medical datasets, where a pure transformer may require more data or stronger regularization than is available.

Recent medical-imaging studies further support the use of ConvNeXt-like architectures. MedNeXt adapts ConvNeXt-style blocks to medical segmentation and emphasizes that convolutional networks can remain competitive in data-scarce medical settings. An improved ConvNeXt-Tiny medical image classification study also shows that ConvNeXt-Tiny can be adapted for efficient medical classification. FocalNeXt, a ConvNeXt-augmented architecture for CT lung cancer classification, provides another example of ConvNeXt components being used in CT classification.

Swin Transformer was included as a comparison because it is a strong hierarchical transformer backbone and its shifted-window attention makes it efficient for images. In this project, Swin-T was a meaningful final comparison model because it represents the transformer family used in many medical-imaging studies. However, the final results showed that ConvNeXt-Tiny was more stable on our limited ROI-based dataset.

The 2.5D strategy is also supported by medical-imaging literature. Several studies use 2.5D representations to preserve partial volumetric context while enabling efficient 2D transfer learning. This is consistent with our classification design, where axial, coronal, and sagittal ROI views are stacked as a three-channel input.

3.4 Modern Techniques and Frameworks in the Field

Modern medical image analysis increasingly combines segmentation and classification. Segmentation provides anatomical localization, while classification uses localized information to predict clinical labels. This is important in tasks where full-image classifiers may learn shortcuts or irrelevant correlations. The proposed system follows this direction by using SwinUNETR for anatomical localization and ConvNeXt-Tiny for classification.

Frameworks such as MONAI and PyTorch support this workflow by providing medical imaging transforms, 3D neural-network components, sliding-window inference, and GPU-based training tools. The classification notebook also uses Torchvision backbones and scikit-learn evaluation metrics. The overall pipeline therefore combines medical-imaging-specific components with general-purpose deep learning libraries.

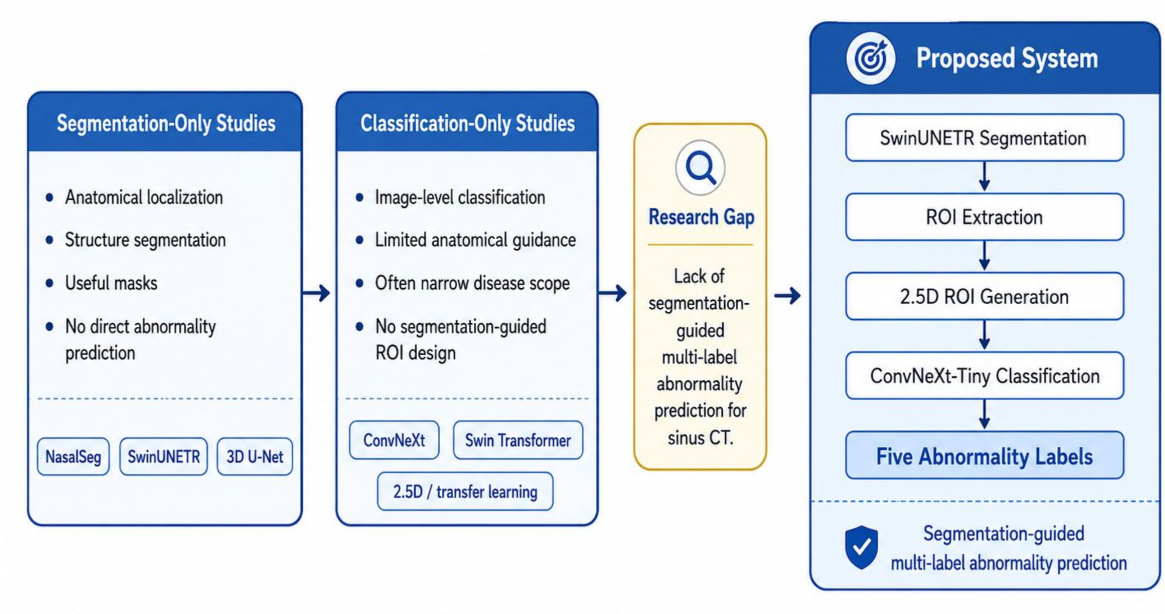

Figure 3.1: Research gap and positioning of the proposed system

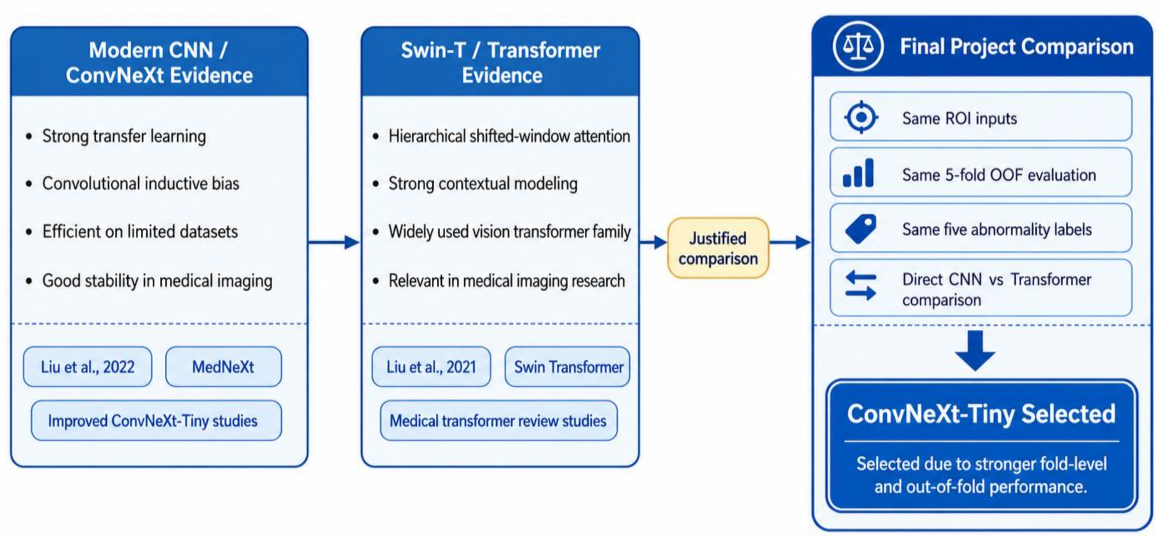

Figure 3.2: Literature-based justification of ConvNeXt-Tiny and Swin-T comparison

Table 3.1: Summary of related work in sinus CT analysis, backbones, and medical imaging

Paper	Approach	Architecture	Main	Strength	Limitation
Zhang et al., 2024, NasalSeg	Open-access CT dataset and benchmark for nasal cavity and	Dataset with voxel-level annotations; benchmark segmentation models.	Provides 130 3D CT scans with five annotated structures.	Public dataset directly related to this project.	Labels only five structures and do not directly include septum or

	paranasal sinus segmentation.				turbinates.
Hatamizadeh et al., 2022, SwinUNETR	Transformer-based 3D medical image segmentation.	Swin Transformer encoder with U-shaped decoder.	Captures long-range context in 3D segmentation.	Strong architecture for volumetric medical segmentation.	Memory intensive and originally validated on other medical tasks.
Liu et al., 2022, ConvNeXt	Modernized convolutional architecture.	Pure ConvNet with transformer-era design improvements.	Shows that modern CNNs can compete with transformer backbones.	Supports choosing ConvNeXt-Tiny for limited-data transfer learning.	Designed for natural images; medical transfer still requires careful ROI design.
Liu et al., 2021, Swin Transformer	Hierarchical transformer with shifted windows.	Window-based self-attention and multi-scale features.	Efficient transformer backbone for vision tasks.	Good comparison model against ConvNeXt.	Can require more data and careful regularization.
Whangbo et al., 2024	Multi-class paranasal sinus segmentation in sinusitis CT.	3D U-Net variants.	Segments major sinus regions in CT images.	Clinically relevant paranasal sinus segmentation setting.	Focuses on segmentation rather than abnormality prediction.
Bhattacharya et al., 2024	Self-supervised learning for paranasal anomaly classification.	3D convolutional autoencoder / self-supervised pipeline.	Classifies normal versus anomalous maxillary sinus patterns.	Addresses limited labeled data through self-supervision.	Narrower classification scope than multi-label abnormality prediction.
Ozturk et al., 2024	Automatic maxillary sinus segmentation on CBCT.	U-Net segmentation model.	Segments maxillary sinus structures.	Relevant to maxillary anatomy and dental/sinus imaging.	CBCT-focused and not a full multi-label CT abnormality pipeline.
Song et al., 2025	Comparison of CNNs, ViTs, and hybrid networks for sinus segmentation.	CNN, ViT, and hybrid segmentation networks.	Compares architecture families for paranasal sinus segmentation.	Useful for understanding architecture trade-offs.	Segmentation-focused and does not solve clinical multi-label classification.

Xia et al., 2025	Medical image classification using an improved ConvNeXt-Tiny.	Lightweight ConvNeXt-Tiny with pooling and attention refinements.	Shows ConvNeXt-Tiny can be adapted to medical classification.	Supports ConvNeXt-Tiny as a reasonable classification backbone.	Not specific to sinus CT or multi-label ROI classification.
Roy et al., 2023, MedNeXt	Medical image architecture based on ConvNeXt-style blocks.	3D ConvNeXt encoder-decoder.	Adapts ConvNeXt principles to medical imaging.	Supports modern CNNs in data-scarce medical settings.	Segmentation-focused, not a direct classifier for sinus CT.
Gulsoy and Kablan, 2025, FocalNeXt	CT lung cancer classification using ConvNeXt-augmented architecture.	ConvNeXt + FocalNet hybrid.	Applies ConvNeXt-related features to CT classification.	Shows ConvNeXt components are useful in CT classification.	Different disease and dataset.
Aburass et al., 2025	Review of vision transformers in medical imaging.	Review of ViT/Swin-style methods.	Summarizes transformer applications in medical classification and segmentation.	Justifies including Swin-T as a comparison.	Review-level evidence; not a sinus-specific experiment.
Wang et al., 2024	2.5D deep learning and conventional features for CT-based prediction.	2.5D masks/features plus model fusion.	Uses 2.5D representation to balance context and efficiency.	Supports 2.5D design for limited data.	Different clinical target and not sinus CT.

3.5 Comparative Analysis of Existing Systems

Table 3.2: Comparison of existing approaches and the proposed system

System / Approach	Features	Technologies	Strengths	Weaknesses
Manual CT interpretation	Radiologist reviews full CT volume and reports findings.	Clinical expertise and PACS viewer.	Gold standard for clinical decision making.	Time-consuming and subjective variability can occur.
Segmentation-only AI	Automatically segments	U-Net, nnU-Net,	Provides localization and	Does not directly predict abnormality

	anatomical structures.	SwinUNETR, 3D CNNs.	quantitative anatomy.	labels.
Image-level classification AI	Predicts disease label from full images or slices.	CNNs, ViTs, transfer learning.	Simple inference pipeline.	May ignore anatomy and learn irrelevant correlations.
Backbone-only transfer learning	Uses pretrained 2D networks on slices or projections.	ConvNeXt, Swin, EfficientNet, ResNet.	Efficient and strong baseline.	Needs careful input design to avoid losing anatomical context.
Proposed system	Segments anatomy, extracts ROIs, predicts five abnormalities.	SwinUNETR + ROI extraction + ConvNeXt-Tiny.	Combines localization and classification.	Limited by small dataset, indirect masks for septum/turbinates, and label imbalance.

3.6 Research Gap and Proposed Innovation

The reviewed work shows that strong methods exist for medical segmentation and image classification, but there remains a gap in segmentation-assisted multi-label abnormality prediction for sinus CT using the available NasalSeg anatomical structures. NasalSeg provides useful masks but does not directly provide all structures required by the abnormality labels, especially the septum and turbinates. Therefore, a direct segmentation-to-label mapping is not possible for all targets.

The proposed innovation is to use the available anatomical masks as stable context regions. The final classifier does not attempt to create brittle artificial masks for unsegmented structures. Instead, it uses clinically motivated neighboring regions: the septum is modeled through the region between both nasal cavities, and inferior turbinate hypertrophy is modeled through the ipsilateral nasal cavity ROI. This is a practical compromise between available segmentation labels and target abnormality labels.

The classification-backbone literature also supports the final comparison between ConvNeXt-Tiny and Swin-T. ConvNeXt-Tiny represents a modern CNN with strong inductive bias and efficient transfer learning, while Swin-T represents the transformer family with hierarchical attention. The project therefore compares both families directly instead of assuming one is better.

3.7 Chapter Summary

This chapter shows that the proposed project is positioned between segmentation-only sinus CT systems and image-level classification systems. Its main novelty is the use of segmentation-guided ROIs to support multi-label abnormality prediction, while explicitly addressing the limitations caused by missing direct septum and turbinate masks. The literature also provides a clear justification for comparing ConvNeXt-Tiny and Swin-T in the final classification phase.

Chapter Four: Research Methodology

4.1 Introduction

This chapter describes the methodology used to develop the AI system for sinus CT analysis. The pipeline begins with CT volume preprocessing and anatomical segmentation. The segmentation output is used to extract fixed-size regions of interest. These ROIs are converted into 2.5D views and passed to a multi-label classifier. The final system predicts five abnormality labels and selects ConvNeXt-Tiny as the final classification model.

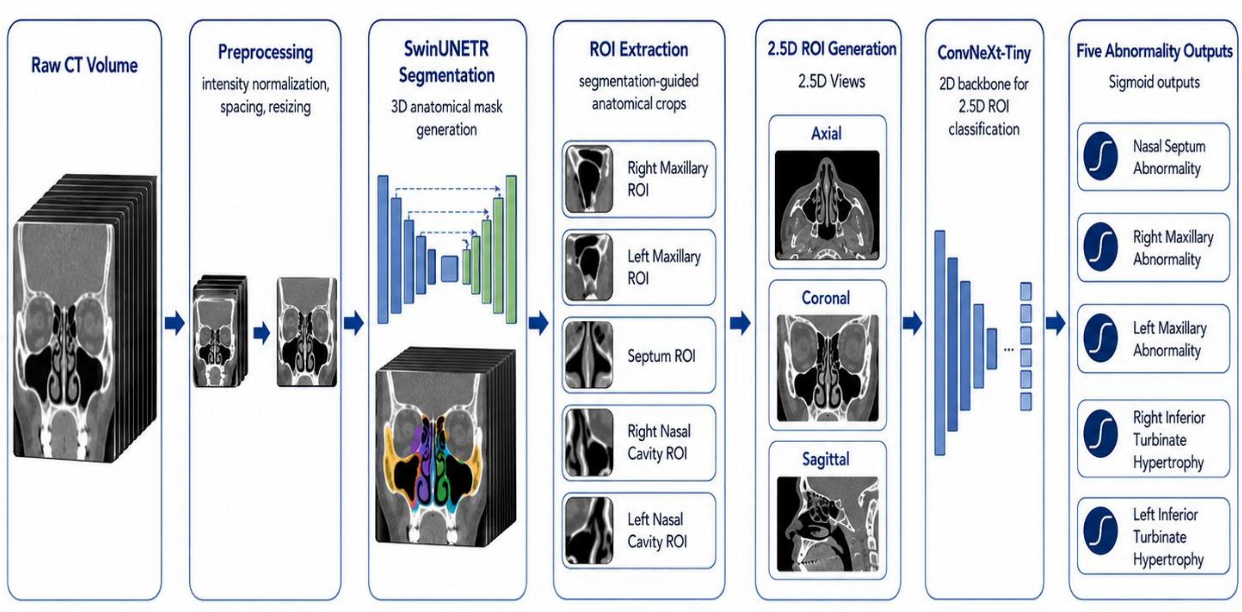

Figure 4.1: Overall pipeline of the proposed sinus CT analysis system

4.2 Dataset Description and Annotation Characteristics

The project uses CT volumes and segmentation labels from the NasalSeg dataset, together with the provided clinical annotation workbook. The segmentation task uses 130 CT cases with five annotated anatomical structures: right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx. The classification task uses the same 130 matched cases and converts selected annotation columns into five binary abnormality targets.

Table 4.1: Dataset summary

Item	Value
Total matched CT cases	130
Normal cases in annotation workbook	18
Cases with at least one selected abnormal label	107

Final classification task	Five-label multi-label abnormality prediction
Segmentation classes	Background + five anatomical structures
Classification evaluation	Balanced 5-fold out-of-fold evaluation

Table 4.2: Final target labels and positive sample counts

Target label	Positive count	Interpretation
nasal_septum_abnormal	71	Any non-empty septum abnormality annotation mapped to positive.
right_maxillary_abnormal	22	Right maxillary sinus abnormality present.
left_maxillary_abnormal	23	Left maxillary sinus abnormality present.
right_inferior_turbinate_hypertrophy	60	Right inferior turbinate hypertrophy present.
left_inferior_turbinate_hypertrophy	60	Left inferior turbinate hypertrophy present.

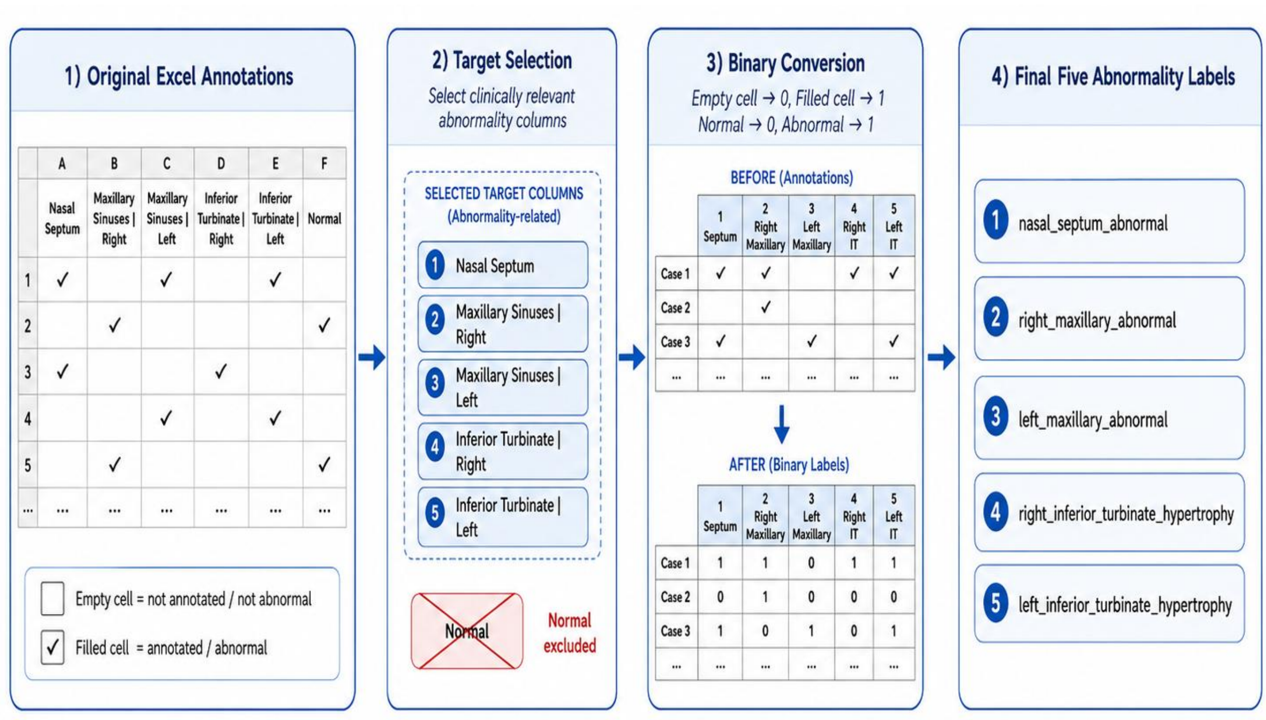

Figure 4.2: Label preparation workflow from the annotation sheet

4.3 Data Preprocessing

For segmentation, CT volumes are loaded as 3D medical images and transformed using MONAI. The main preprocessing operations include orientation handling, spacing normalization, intensity windowing, scaling, and conversion to tensors. The HU window used in the segmentation notebook is -400 to 1000, which keeps relevant sinonasal soft tissue and bony structures while limiting extreme intensity values.

For classification, each CT and segmentation pair is used to extract anatomy-guided ROIs. Each ROI includes two channels: the CT intensity crop and the corresponding segmentation or mask context. The ROI crop is resized to a fixed size, then converted into 2.5D images by taking representative axial, coronal, and sagittal planes. These three planes form a three-channel image suitable for pretrained 2D backbones.

4.4 Data Augmentation

Data augmentation was applied only to the training data. For segmentation, augmentation included random crops, flips, rotations, Gaussian noise, and intensity shift or scale transformations. For classification, augmentation was applied to improve robustness of the 2.5D ROI views without changing the target labels. Validation and test samples were kept stable except for deterministic preprocessing so that evaluation remained unbiased.

4.5 Data Partitioning and 5-Fold Evaluation

The segmentation notebook used an executed split of 91 training cases, 19 validation cases, and 20 test cases. The classification notebook used balanced 5-fold out-of-fold evaluation across the 130 matched cases. This setup ensured that each case was evaluated by a model that did not train on it. The 5-fold design was selected because the dataset is limited and a single train/test split could produce unstable results, especially for rare labels such as maxillary abnormalities.

Table 4.3: Data partitioning strategy used in this project

Stage	Partitioning	Purpose
Segmentation	91 train / 19 validation / 20 test	Train SwinUNETR and evaluate anatomical segmentation on held-out cases.
Classification	Balanced 5-fold OOF	Compare classifiers while using every case once as validation.
Threshold tuning	OOF predictions	Select label-specific probability thresholds for final metrics.

4.6 SwinUNETR Segmentation Stage

The segmentation stage uses SwinUNETR, a 3D transformer-based U-shaped architecture. The model is trained to predict six classes: background, right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx. The training objective is DiceCE loss, which combines overlap-based Dice loss and cross-entropy loss. This is suitable for segmentation because it encourages both voxel-level class prediction and mask overlap.

Table 4.4: SwinUNETR segmentation training configuration

Parameter	Value
Architecture	SwinUNETR
Framework	MONAI / PyTorch
Input spacing	1.0 × 1.0 × 1.5
ROI size	96 × 96 × 32
HU window	-400 to 1000
Loss function	DiceCE loss
Optimizer	AdamW
Validation method	Sliding-window inference
Executed split	91 train / 19 validation / 20 test cases

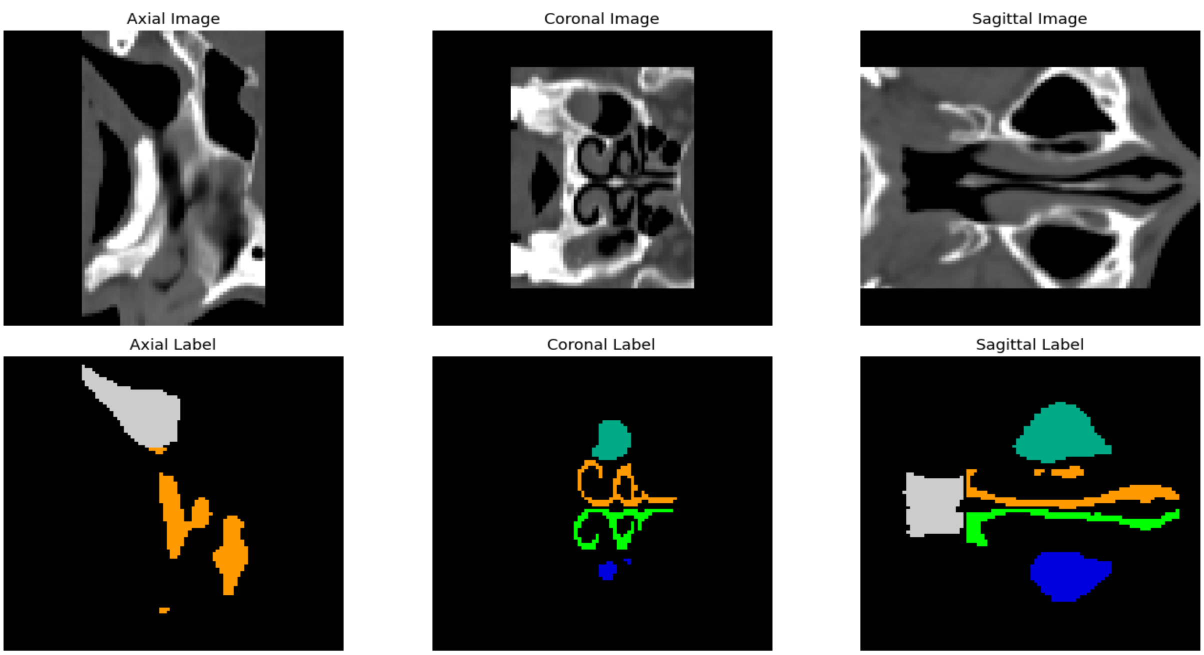

Figure 4.3: Augmentation sanity-check visualization

4.7 ROI Extraction Strategy

The final classification model uses five stable contextual ROIs. The right and left maxillary labels use the corresponding maxillary sinus masks directly. The nasal septum label uses a contextual ROI derived from both nasal cavities with padding. This design follows the anatomical fact that the nasal septum lies between the two nasal cavities, while avoiding brittle pseudo-septum masks. The inferior turbinate labels use the ipsilateral nasal cavity ROIs because direct turbinate masks are not available in the dataset.

Table 4.5: ROI-to-label mapping used in the final classification system

Target label	ROI name	Segmentation labels used	Reasoning
right_maxillary_abnormal	right_maxillary_roi	Right maxillary	Direct

		sinus	anatomical match.
left_maxillary_abnormal	left_maxillary_roi	Left maxillary sinus	Direct anatomical match.
nasal_septum_abnormal	nasal_septum_region_roi	Right + left nasal cavities	Septum lies between both nasal cavities; contextual ROI is more stable.
right_inferior_turbinate_abnormal	right_nasal_cavity_roi	Right nasal cavity	Inferior turbinate is located inside the nasal cavity; no direct turbinate mask available.
left_inferior_turbinate_abnormal	left_nasal_cavity_roi	Left nasal cavity	Inferior turbinate is located inside the nasal cavity; no direct turbinate mask available.

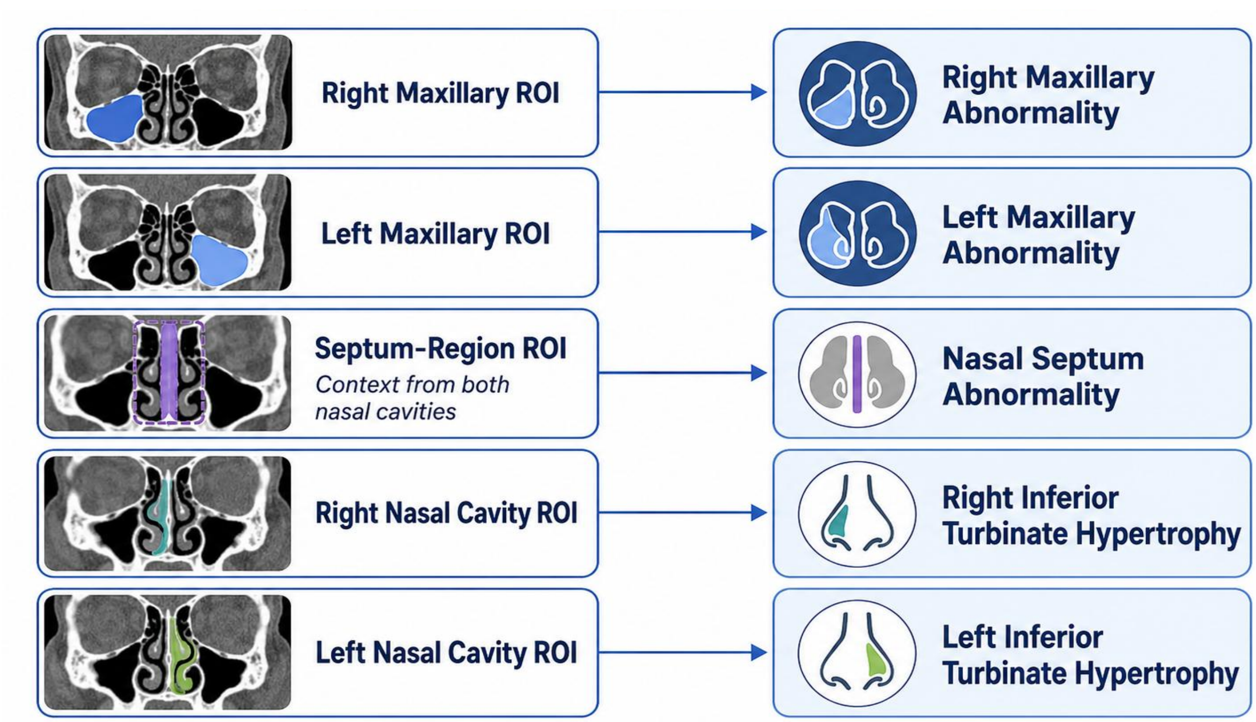

Figure 4.4: Final ROI definitions used in the classification stage

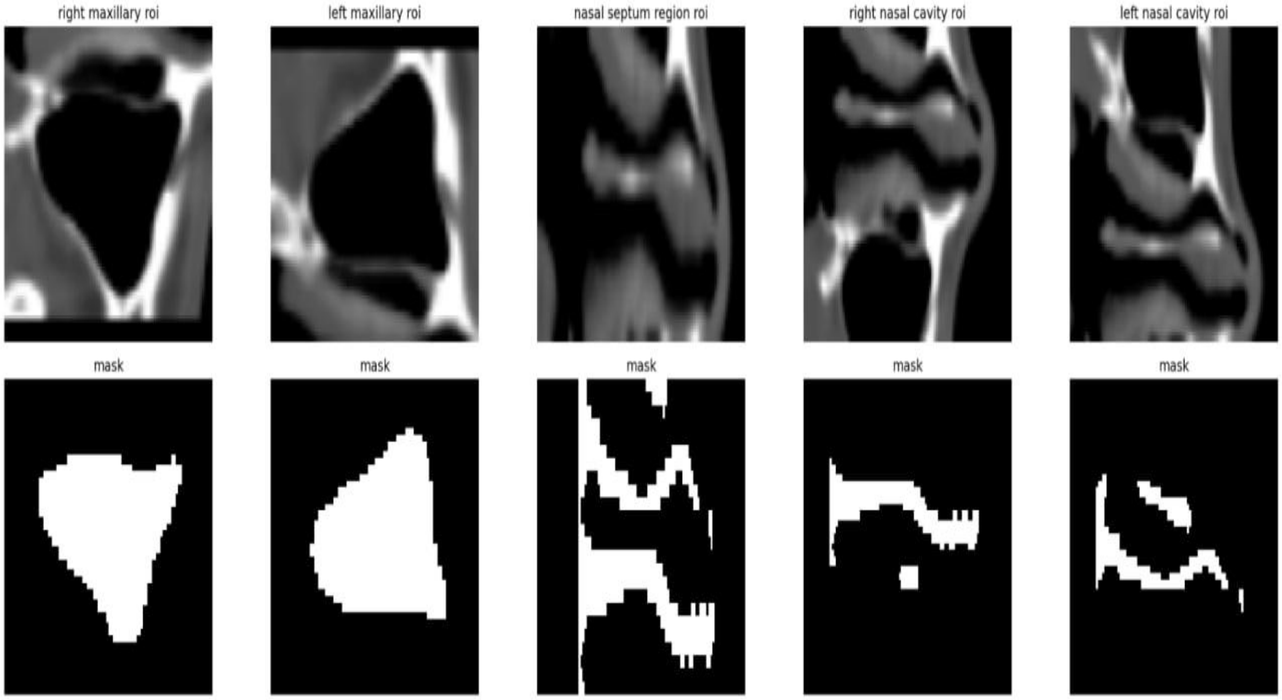

Figure 4.5: Example ROI extraction results from the classification notebook

4.8 2.5D ROI Generation

Each 3D ROI was converted into a 2.5D representation by taking representative axial, coronal, and sagittal views. These three views were stacked as a three-channel image. This design preserved partial volumetric context while allowing the use of pretrained 2D image classification backbones. It also reduced the memory cost compared with fully 3D classifiers, which is important for a dataset of 130 cases.

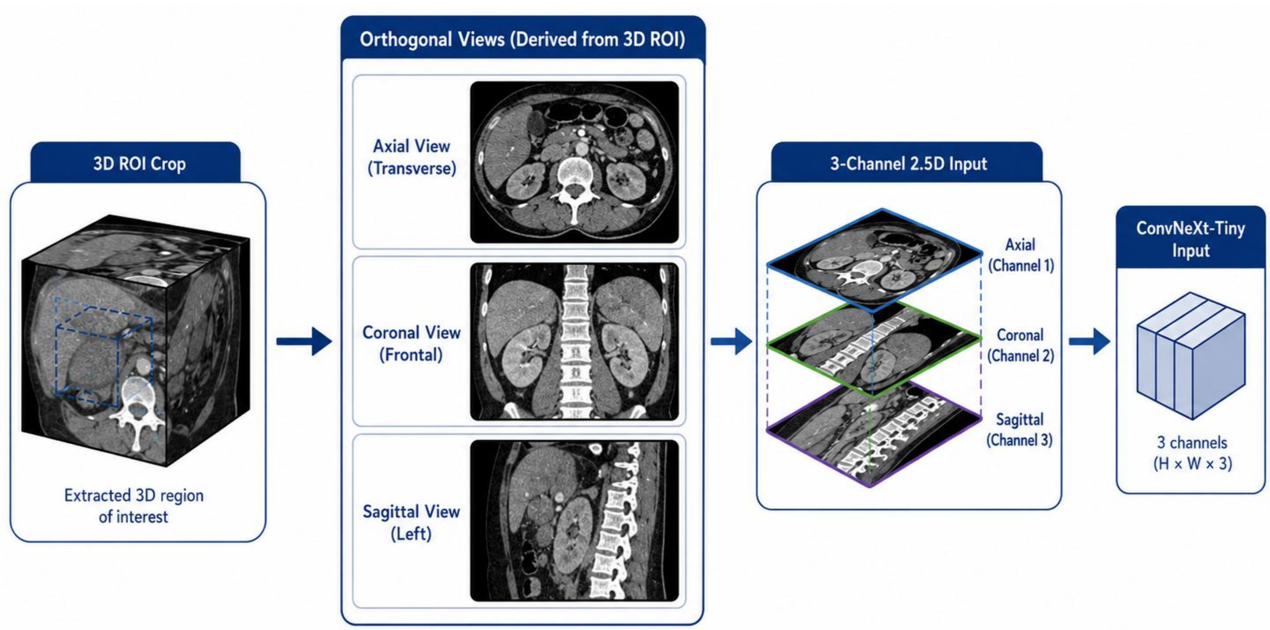

Figure 4.6: 2.5D ROI generation from axial, coronal, and sagittal views

4.9 Important Exploratory Experiments

The project evolved through a sequence of major experiments. The purpose of this section is to document the important development path in an academic way rather than listing every minor run. The progression shows how the system moved from a segmentation foundation and early unstable classification attempts toward the final ConvNeXt-Tiny model.

Segmentation foundation: The first major stage was training the SwinUNETR segmentation model on NasalSeg. This stage was necessary because all later classification experiments depended on segmentation-guided ROI extraction.

Early 3D ROI classification: After segmentation, the first classification direction used extracted anatomical crops with ROI-level classifiers, including a custom 3D ROI ResNet18-style classifier. This established the idea of region-specific prediction, but it was limited by dataset size and 3D training instability.

2.5D transfer-learning transition: To improve stability with limited data, the pipeline moved to 2.5D transfer learning. Each ROI was converted into axial, coronal, and sagittal views so that pretrained backbones could be used.

Normal label removal: The Normal column was not used as a sixth target because the project predicts abnormalities. Cases without selected abnormalities are naturally represented by all-zero target vectors.

Septum ROI experiments: A narrow pseudo-septum region was considered by deriving a region between the right and left nasal cavities. This was brittle because NasalSeg does not directly annotate the septum. A broader nasal-cavity context was more stable, leading to the final septum-region ROI built from both nasal cavities with padding.

Turbinate ROI experiments: Direct pseudo-turbinate crops were considered, but they did not improve results reliably because NasalSeg does not provide turbinate masks. The final design predicts inferior turbinate hypertrophy from the ipsilateral nasal cavity ROI.

Imbalance handling: Several imbalance strategies were considered. The final serious experiments used binary focal loss with gamma equal to 1 and positive class weighting. Random weighted sampling was not retained in the final configuration because the selected setup was more stable.

Final backbone comparison: The final classification comparison focused on ConvNeXt-Tiny and Swin-T under the same ROI cache, targets, and 5-fold evaluation. ConvNeXt-Tiny achieved the strongest overall performance and was selected as the final model.

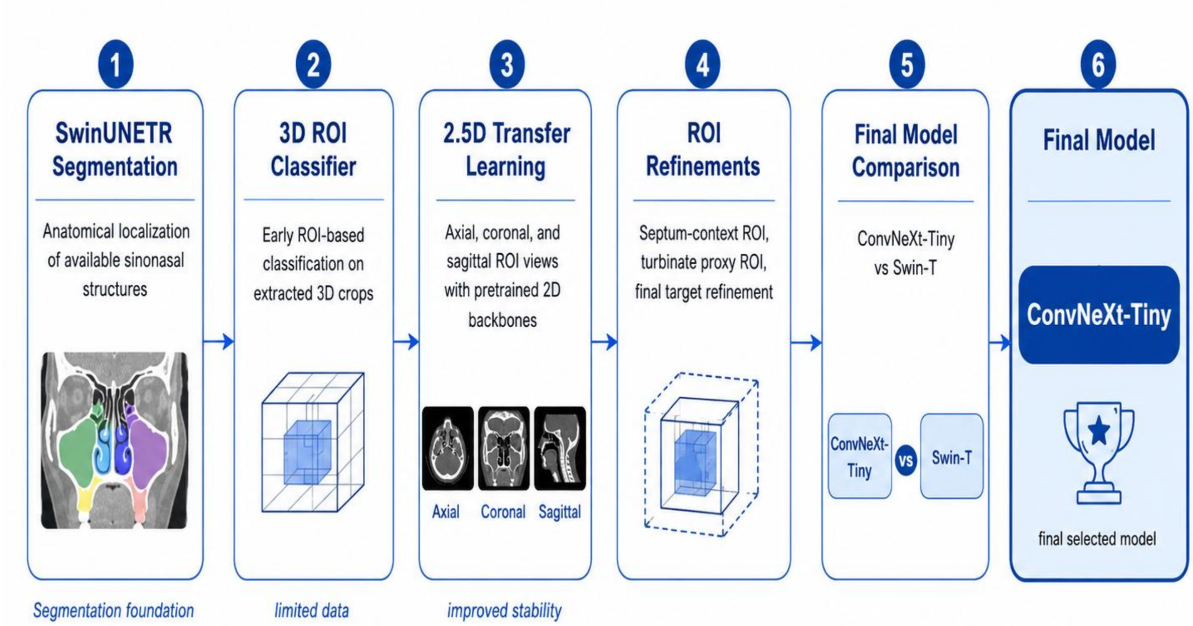

Figure 4.7: Timeline of major classification experiments

Table 4.6: Timeline of major exploratory experiments and outcomes

Stage	Experiment / Idea	Main observation	Decision
1	SwinUNETR segmentation	Provided reliable anatomical localization for available NasalSeg structures.	Kept as segmentation foundation.
2	3D ROI ResNet-style classifier	Region-specific prediction was reasonable but unstable with limited data.	Moved to 2.5D transfer learning.
3	2.5D transfer learning	Allowed use of pretrained backbones and improved stability.	Adopted for final classification.
4	Normal as target	Not aligned with abnormality prediction and multi-label interpretation.	Ignored in final target set.
5	Pure septum pseudo-ROI	Too brittle because septum is not directly segmented.	Not used.
6	Contextual septum ROI	More stable and anatomically meaningful.	Adopted.
7	Turbinate pseudo-ROI	Unstable due to lack of direct turbinate masks.	Use ipsilateral nasal cavity ROI.
8	ConvNeXt-Tiny vs Swin-T	ConvNeXt-Tiny achieved better macro F1 and exact match.	ConvNeXt-Tiny selected.

4.10 Final Classification Models

The classification stage compares two final 2.5D transfer-learning models: ConvNeXt-Tiny and Swin-T. Both models use the same target labels, ROI cache, augmentation strategy, and threshold-tuned out-of-fold evaluation. Each label is predicted using a sigmoid output. The training uses binary focal loss with gamma equal to 1, positive class weighting, and 5-fold validation.

Table 4.7: Final classification experiment settings

Setting	ConvNeXt-Tiny	Swin-T
Experiment name	01_final_convnext_tiny_focal_g1_lr1e4_strict	02_final_swin_t_focal_g1_diff_lr
Backbone	ConvNeXt-Tiny	Swin-T
Model type	2.5D transfer learning	2.5D transfer learning
Loss	Binary focal loss	Binary focal loss
Focal gamma	1.0	1.0
Positive class weighting	Enabled	Enabled
Augmentation	Enabled	Enabled
Head learning rate	1e-4	1e-4
Backbone learning rate	1e-4	1e-5
Dropout	0.35	0.35
Evaluation	5-fold OOF	5-fold OOF

4.11 Implementation Environment

The experiments were implemented in a notebook-based Python environment. The segmentation pipeline used PyTorch and MONAI for 3D medical image processing, transforms, training, and sliding-window inference. The classification pipeline used PyTorch, TorchVision, pandas, NumPy, scikit-learn, and Matplotlib. Training was performed in a Kaggle GPU environment. The final notebooks produced checkpoint files, ROI caches, OOF prediction tables, threshold tables, metrics CSV files, and plots used for analysis.

4.12 Chapter Summary and Research Contribution

The methodology combines 3D anatomical segmentation with ROI-based 2.5D multi-label classification. The major methodological decision is to avoid using the full CT volume as an unlocalized classifier input and instead use segmentation-guided anatomical crops. This makes the pipeline more interpretable and allows each output label to be associated with a clinically meaningful region. The final classifier is ConvNeXt-Tiny, with Swin-T used as the main comparison model.

Chapter Five: Research Results

5.1 Introduction

This chapter presents the experimental results of the proposed sinus CT analysis system. The results are reported in two main parts: segmentation results and classification results. Segmentation results evaluate the SwinUNETR model as the anatomical localization stage. Classification results evaluate the final multi-label abnormality prediction stage and compare ConvNeXt-Tiny with Swin-T.

5.2 Tools and Evaluation Methods

The experiments were implemented using Python, PyTorch, MONAI, Torchvision, NumPy, pandas, scikit-learn, and Matplotlib. Training was performed in a Kaggle GPU environment. The segmentation model was evaluated with Dice, IoU, precision, recall, inference time, and throughput. The classification models were evaluated with macro accuracy, macro precision, macro recall, macro specificity, macro balanced accuracy, macro F1, macro AUC, macro average precision, overall label accuracy, and exact match accuracy.

5.2.1 Experimental Environment

The segmentation and classification experiments were developed as notebook-based workflows. This allowed step-by-step inspection of data loading, preprocessing, ROI extraction, model training, metric computation, and figure generation. The final artifacts included trained checkpoints, ROI cache files, fold summaries, out-of-fold predictions, tuned thresholds, per-label metrics, and plots.

5.2.2 Evaluation Metrics Used in the Final Experiments

Segmentation was evaluated using mean Dice, mean IoU, precision, recall, inference time, and throughput. Classification was evaluated using label-wise and macro metrics. Because the task is multi-label and imbalanced, macro F1 and out-of-fold metrics were emphasized over simple accuracy. Exact match accuracy was also reported, but it is naturally strict because it requires all five labels of each case to be correct at the same time.

5.3 Segmentation Results

The SwinUNETR model achieved strong segmentation performance on the test set. The mean Dice score was 0.8743 and the mean IoU was 0.8102. The high recall value indicates that the model generally captured most of the reference anatomical regions. The segmentation results are suitable for the intended purpose of guiding ROI extraction in the classification stage.

5.3.1 Quantitative Segmentation Results

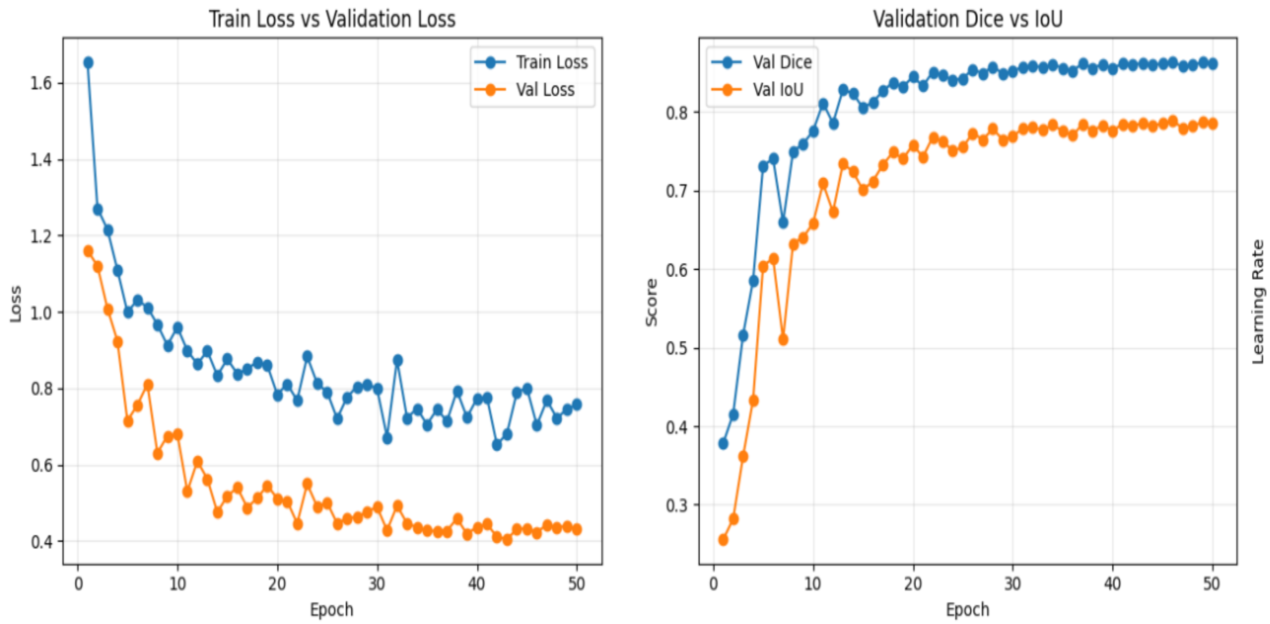

Figure 5.1: SwinUNETR training and validation curves

Table 5.1: Overall segmentation performance on the test set

Metric	Value
Mean Dice	0.8743
Mean IoU	0.8102
Mean Precision	0.8424
Mean Recall	0.9158
Inference time	0.34 s / volume
Throughput	2.90 volumes / s

Table 5.2: Per-class segmentation performance

Class	Dice	IoU	Precision	Recall
Background	0.9896	0.9796	0.9946	0.9846
Right maxillary sinus	0.9107	0.8543	0.8797	0.9476
Left maxillary sinus	0.8697	0.8104	0.8503	0.9053
Right nasal cavity	0.8009	0.7010	0.7440	0.8729
Left nasal cavity	0.8079	0.7138	0.7520	0.8772
Nasopharynx	0.8669	0.8024	0.8338	0.9075

5.3.2 Qualitative Segmentation Results

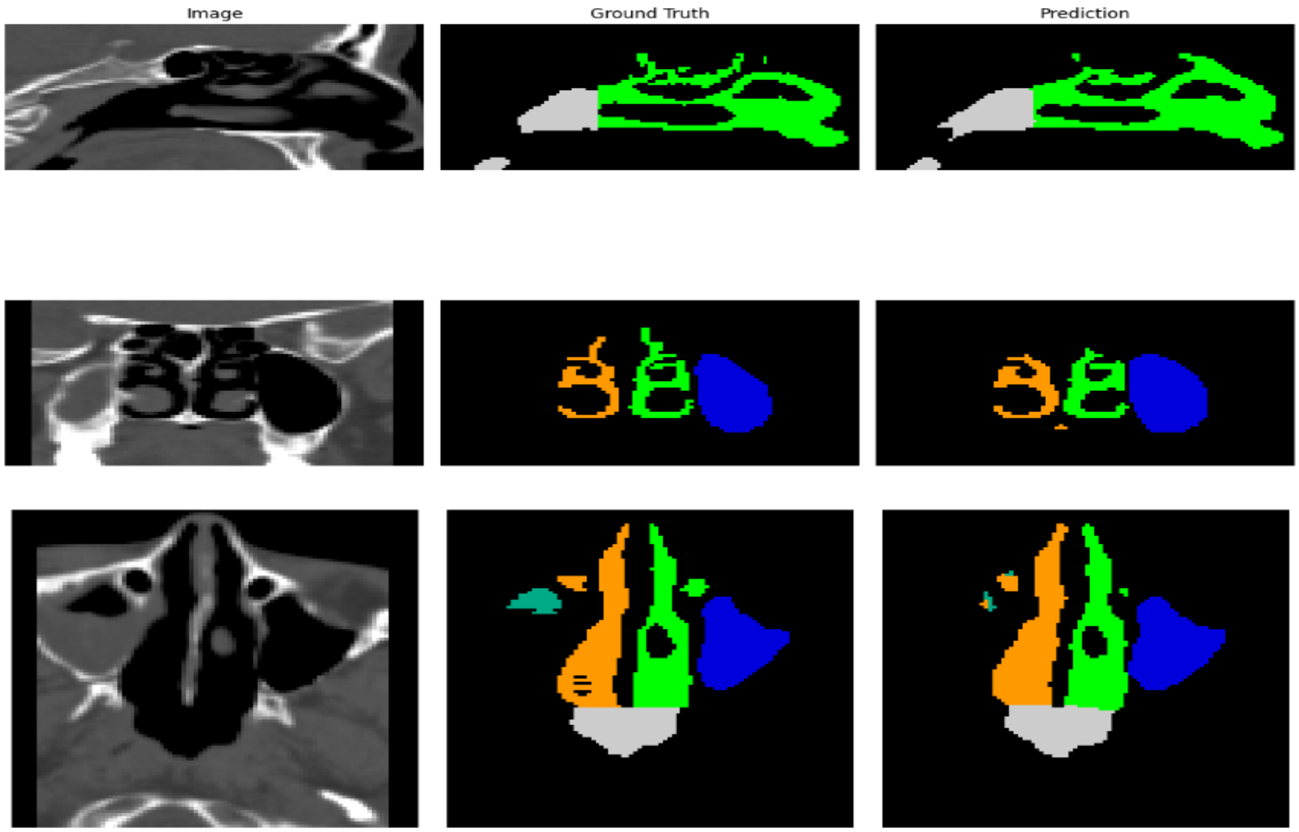

Figure 5.2: Example segmentation prediction on a test case

Qualitative examples should be used to confirm that the segmentation mask follows the expected anatomical structures and that failure cases are not hidden by global metrics. These images will be inserted later from the notebook outputs or manually prepared visualizations.

5.4 Classification Results

The final classification experiments compare ConvNeXt-Tiny and Swin-T using the same 5-fold out-of-fold evaluation setup. ConvNeXt-Tiny achieved higher performance in the main comparison metrics and is therefore selected as the final proposed model. In particular, ConvNeXt-Tiny achieved a fold-level macro F1 of 0.7849 and an out-of-fold macro F1 of 0.6928, compared with 0.7401 and 0.6565 for Swin-T.

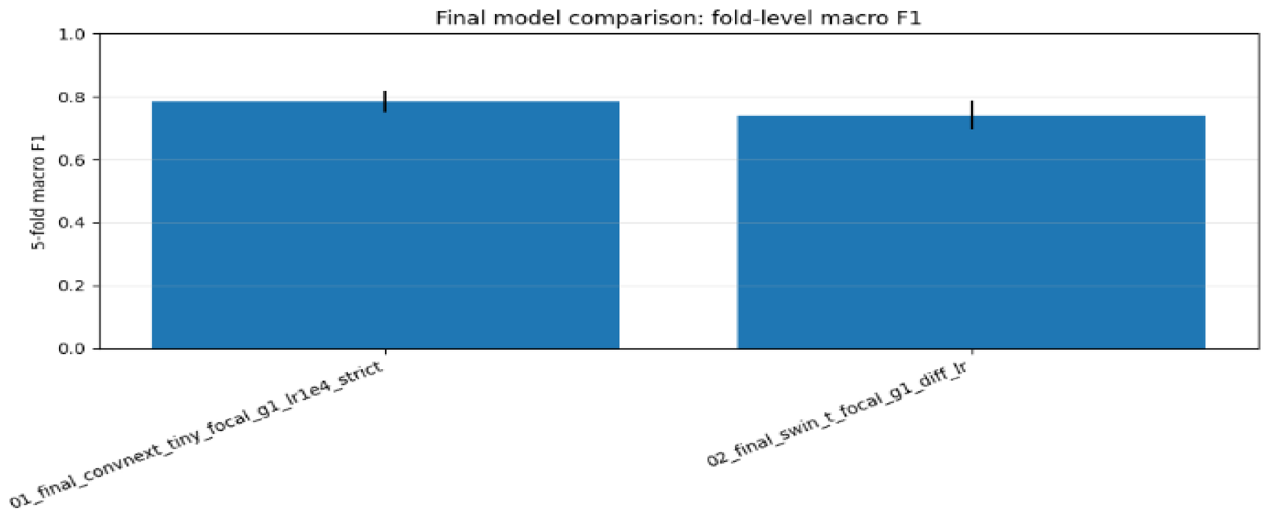

Figure 5.3: Fold-level macro F1 comparison between final models

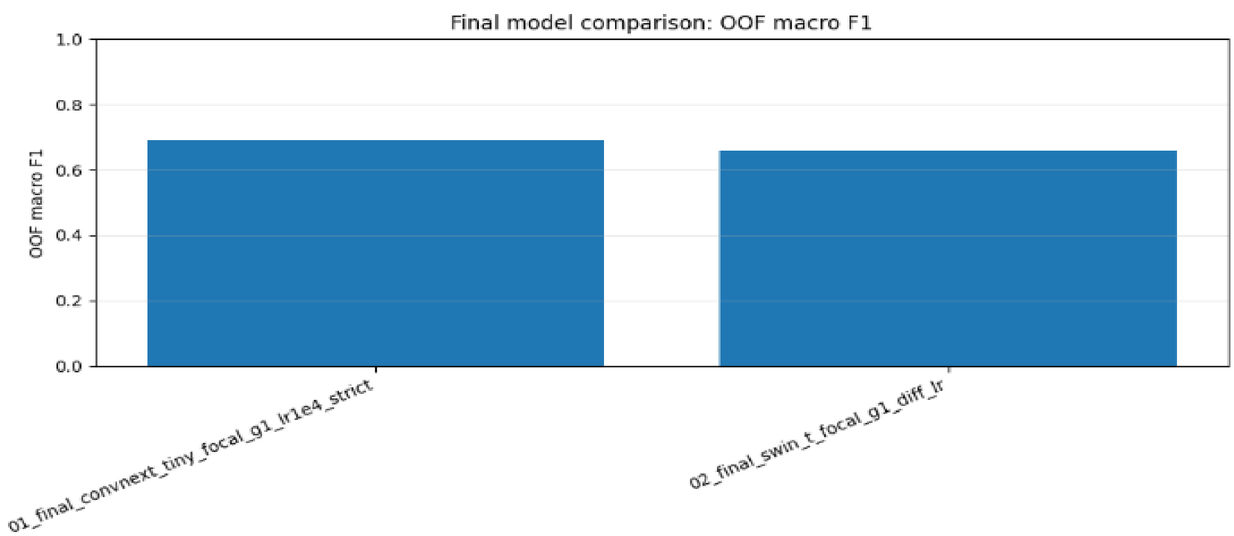

Figure 5.4: OOF macro F1 comparison between final models

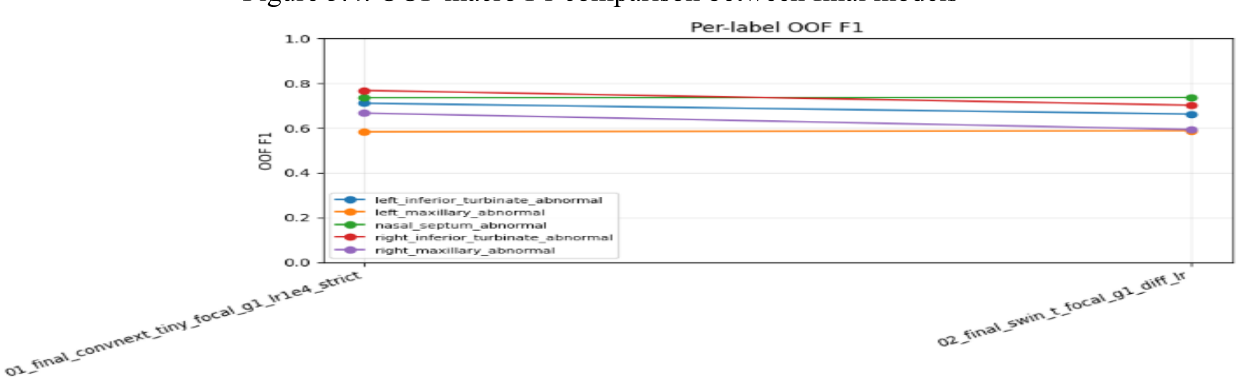

Figure 5.5: Per-label OOF F1 comparison for final models

Table 5.3: Final comparison of ConvNeXt-Tiny and Swin-T

Model	Fold-level macro F1	OOF macro F1	Overall label accuracy	Exact match accuracy
ConvNeXt-Tiny	0.7849	0.6928	0.7877	0.3846
Swin-T	0.7401	0.6565	0.7031	0.1923

5.4.1 Final ConvNeXt-Tiny Results

ConvNeXt-Tiny achieved the best overall classification performance. Its OOF macro F1 of 0.6928 is the main final score because every case is evaluated by a model that did not train on it. Its overall label accuracy was 0.7877, and exact match accuracy was 0.3846. Exact match is strict in this multi-label task because a case with four correct labels and one wrong label is still considered incorrect under this metric.

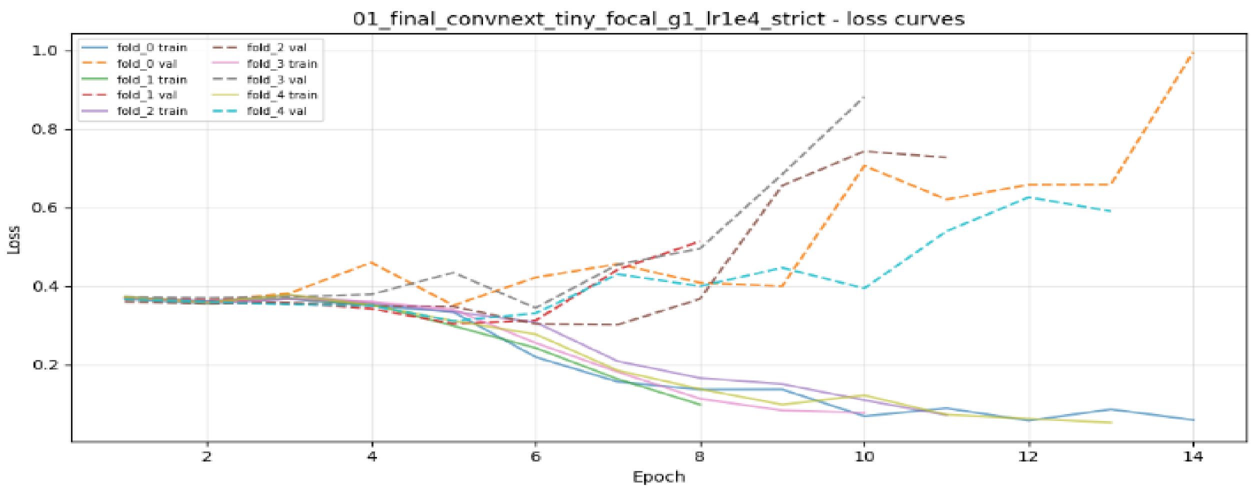

Figure 5.6: ConvNeXt-Tiny loss curves

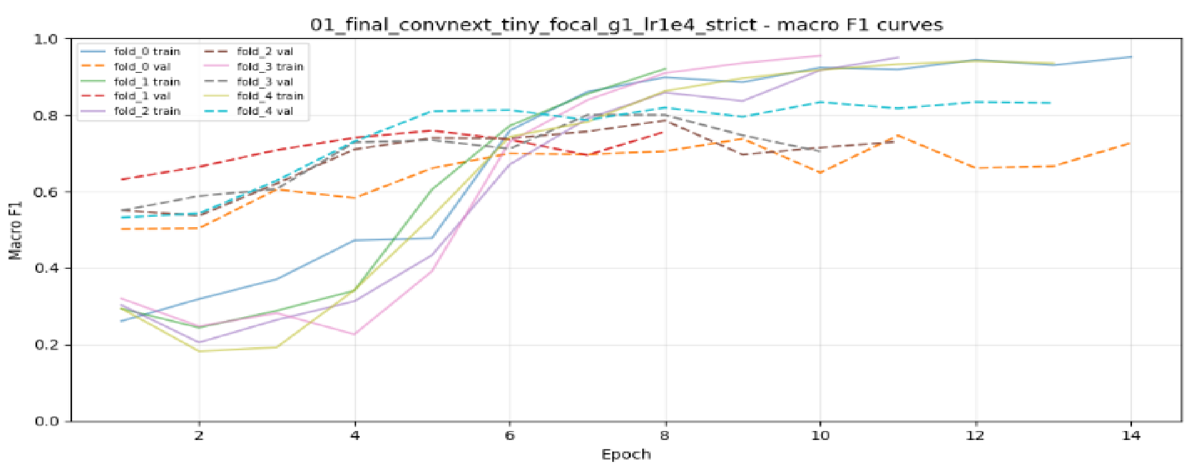

Figure 5.7: ConvNeXt-Tiny macro F1 curves

Table 5.4: OOF per-label metrics for ConvNeXt-Tiny

Label	Threshold	Accuracy	Precision	Recall	F1
left_inferior_turbinate_abnormal	0.49	0.7308	0.7049	0.7167	0.7107
left_maxillary_abnormal	0.42	0.8462	0.5600	0.6087	0.5833
nasal_septum_abnormal	0.44	0.6846	0.6786	0.8028	0.7355
right_inferior_turbinate_abnormal	0.46	0.7769	0.7385	0.8000	0.7680
right_maxillary_abnormal	0.43	0.9000	0.7647	0.5909	0.6667

5.4.2 Swin-T Comparison Results

Swin-T was included as the final transformer comparison model. It achieved competitive results on some labels, especially nasal septum abnormality, but its overall OOF macro F1 and exact match accuracy were lower than ConvNeXt-Tiny. This suggests that, for the available dataset size and ROI-based inputs, the convolutional inductive bias of ConvNeXt-Tiny was more stable.

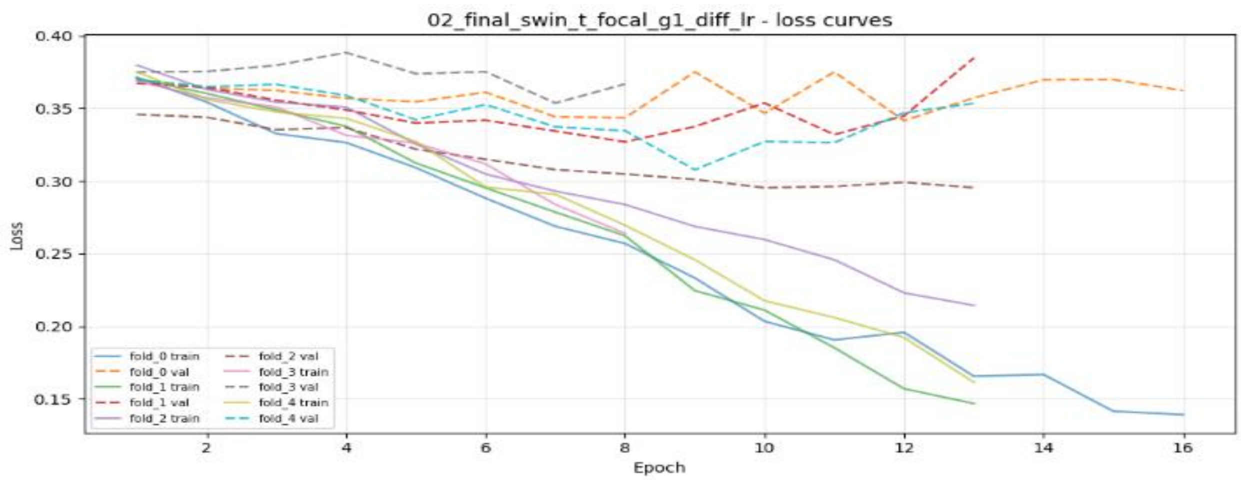

Figure 5.8: Swin-T loss curves

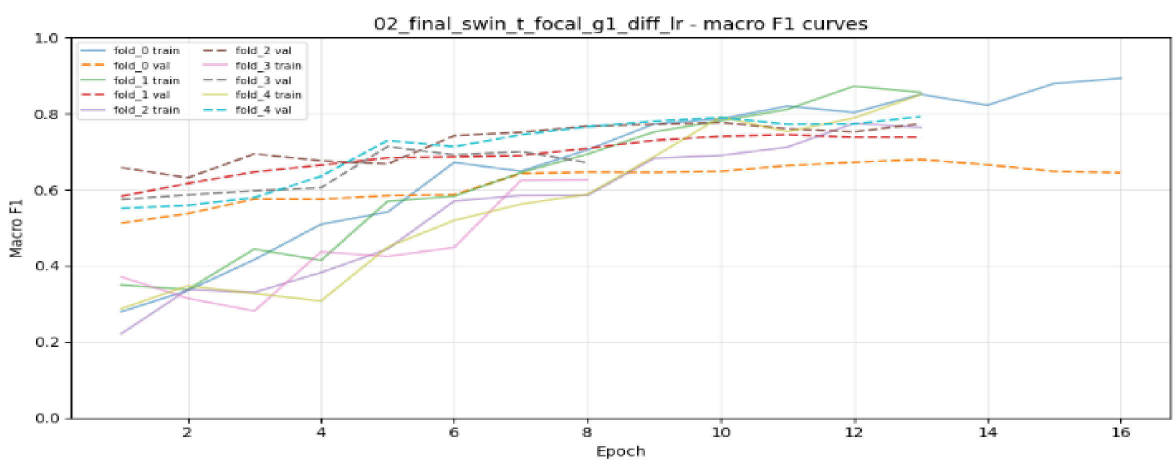

Figure 5.9: Swin-T macro F1 curves

Table 5.5: OOF per-label F1 for Swin-T

Label	F1
left_inferior_turbinate_abnormal	0.6624
left_maxillary_abnormal	0.5882
nasal_septum_abnormal	0.7363
right_inferior_turbinate_abnormal	0.7020
right_maxillary_abnormal	0.5938

5.4.3 Per-label Performance Analysis

The per-label results show that the model performed better on some anatomical targets than others. Right inferior turbinate hypertrophy and nasal septum abnormality achieved relatively strong F1 scores. The maxillary labels were less stable, especially left maxillary abnormality. This does not mean that the ROI design was incorrect; rather, it reflects the small positive counts and frequent co-occurrence of maxillary abnormalities with septal or turbinate findings.

5.4.4 Confusion Matrix and Multi-Label Interpretation

The final notebooks include 5×5 dominant-label confusion matrices for ConvNeXt-Tiny and Swin-T. These matrices are useful for visualizing the dominant predicted abnormality among abnormal cases, but they must be interpreted carefully. The classification task is multi-label, while the matrix forces each abnormal case into a single dominant true label and a single dominant predicted label. Therefore, a case can be partially correct in the multi-label sense but still appear as a confusion in the dominant-label matrix.

This issue is especially important for maxillary abnormalities. Maxillary-positive cases are relatively rare and often co-occur with septal or turbinate abnormalities. As a result, if the model correctly predicts a maxillary label but assigns higher confidence to a co-occurring septal or turbinate label, the dominant-label matrix may underrepresent the true multi-label correctness. For this reason, the report emphasizes macro F1, per-label metrics, and OOF evaluation in addition to the 5×5 visualization.

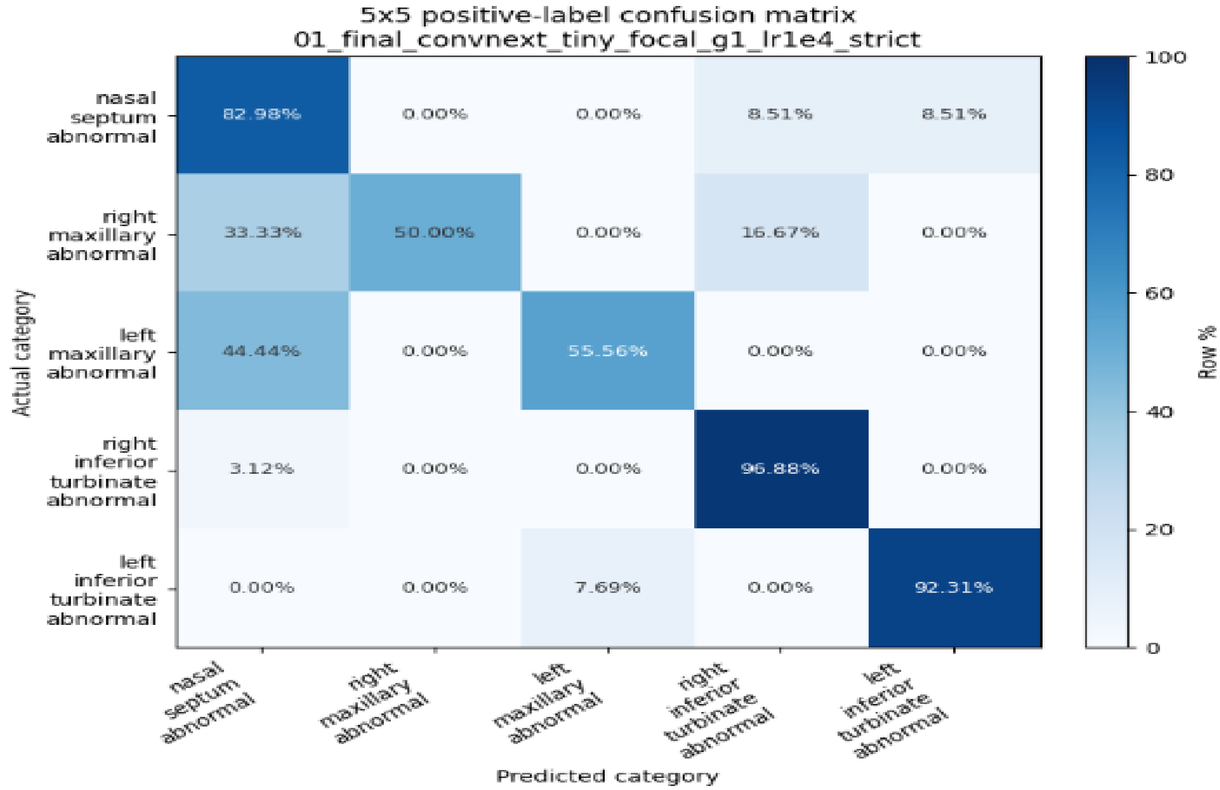

Figure 5.10: ConvNeXt-Tiny 5×5 dominant-label confusion matrix

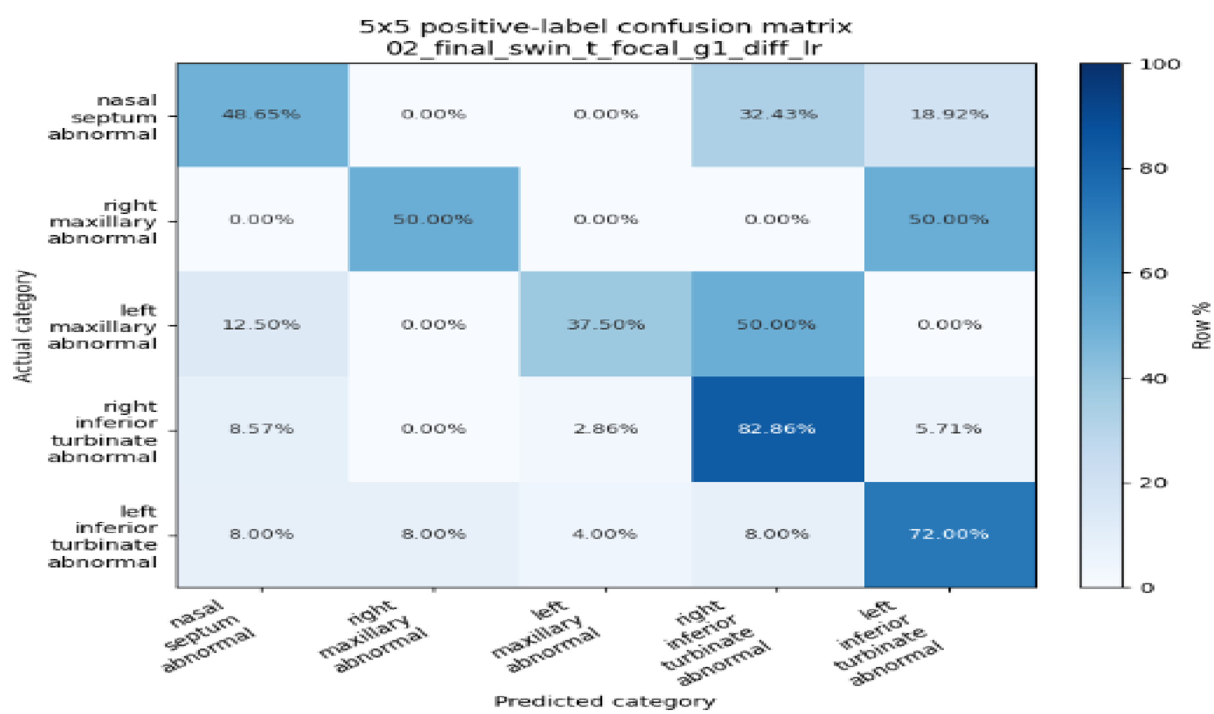

Figure 5.11: Swin-T 5×5 dominant-label confusion matrix

5.4.5 ROC and Precision-Recall Analysis

ROC and precision-recall curves provide additional insight into threshold-dependent behavior. ROC curves show how sensitivity and specificity trade off across probability thresholds. Precision-recall curves are especially important for imbalanced labels because they focus on positive-class retrieval. ConvNeXt-Tiny showed useful separability for several labels, although maxillary labels remained harder because of lower positive counts.

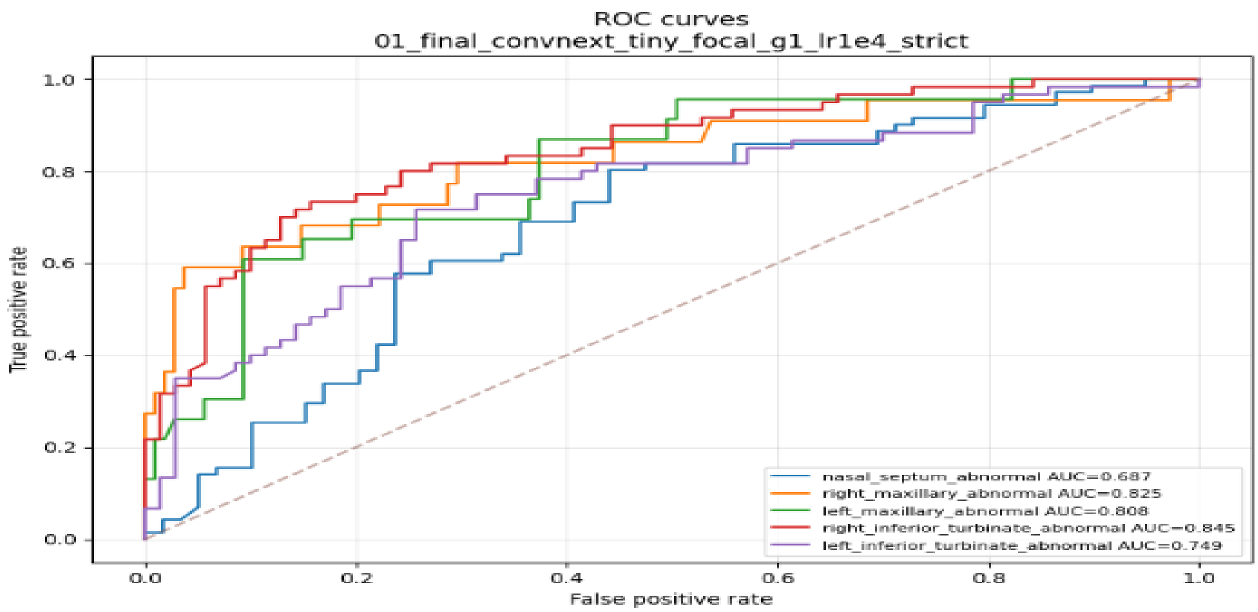

Figure 5.12: ROC curves for the final ConvNeXt-Tiny model

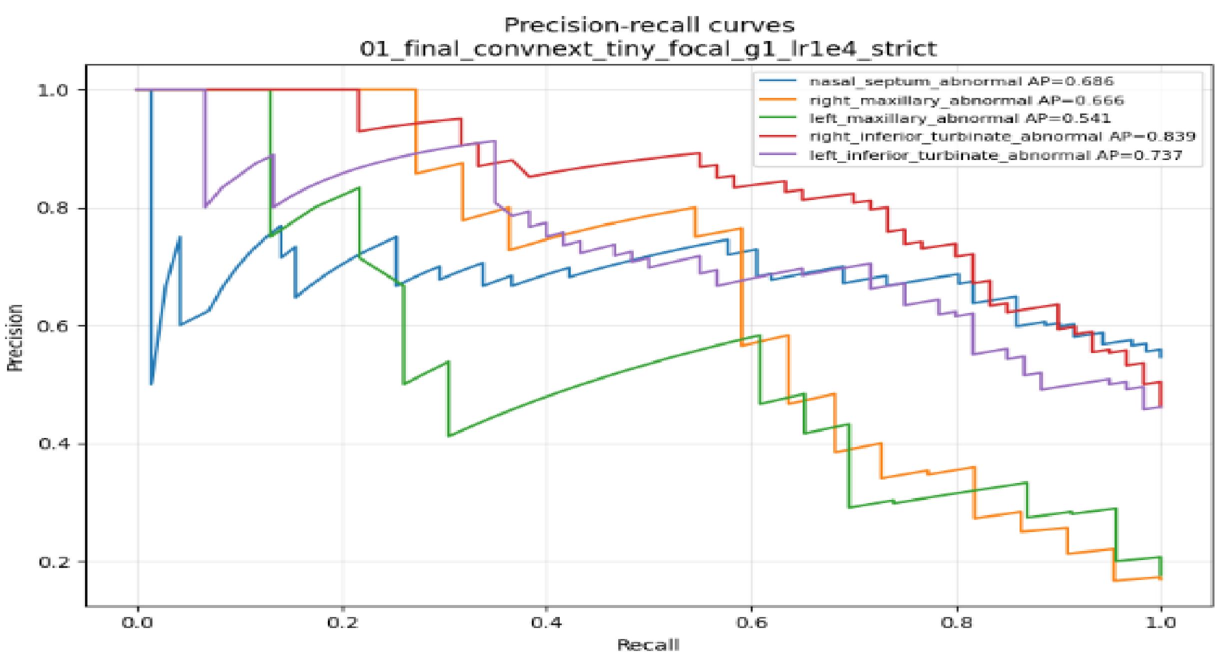

Figure 5.13: Precision-Recall curves for the final ConvNeXt-Tiny model

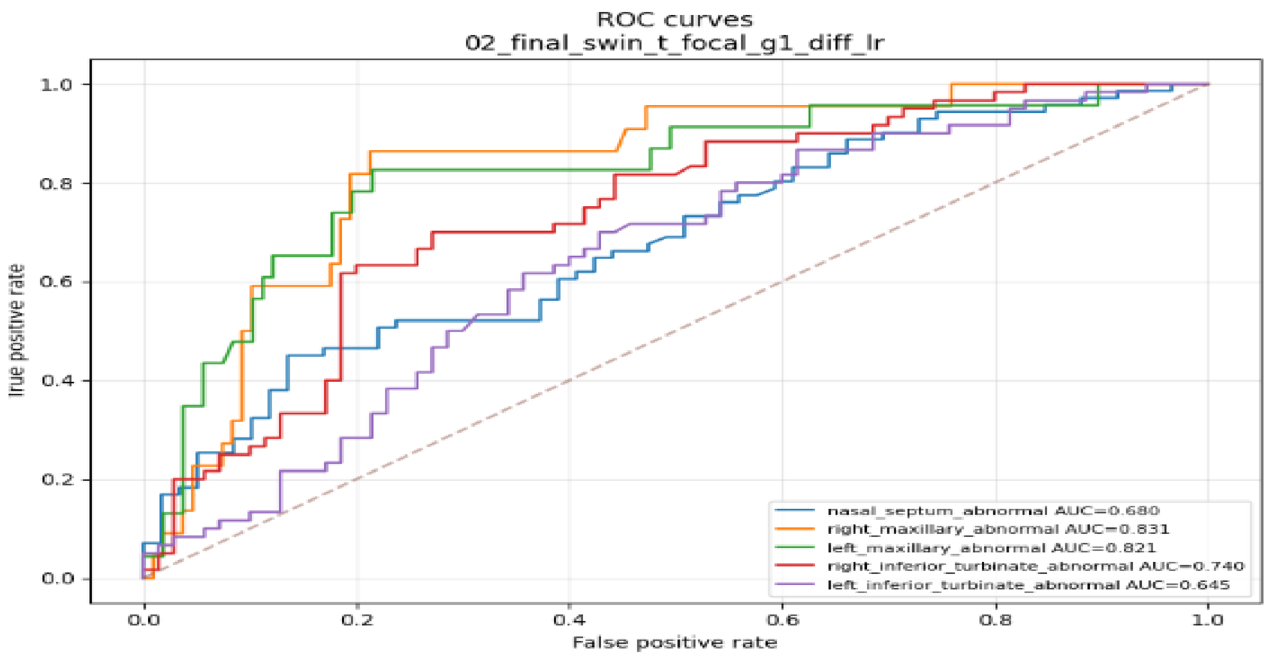

Figure 5.14: ROC curves for Swin-T

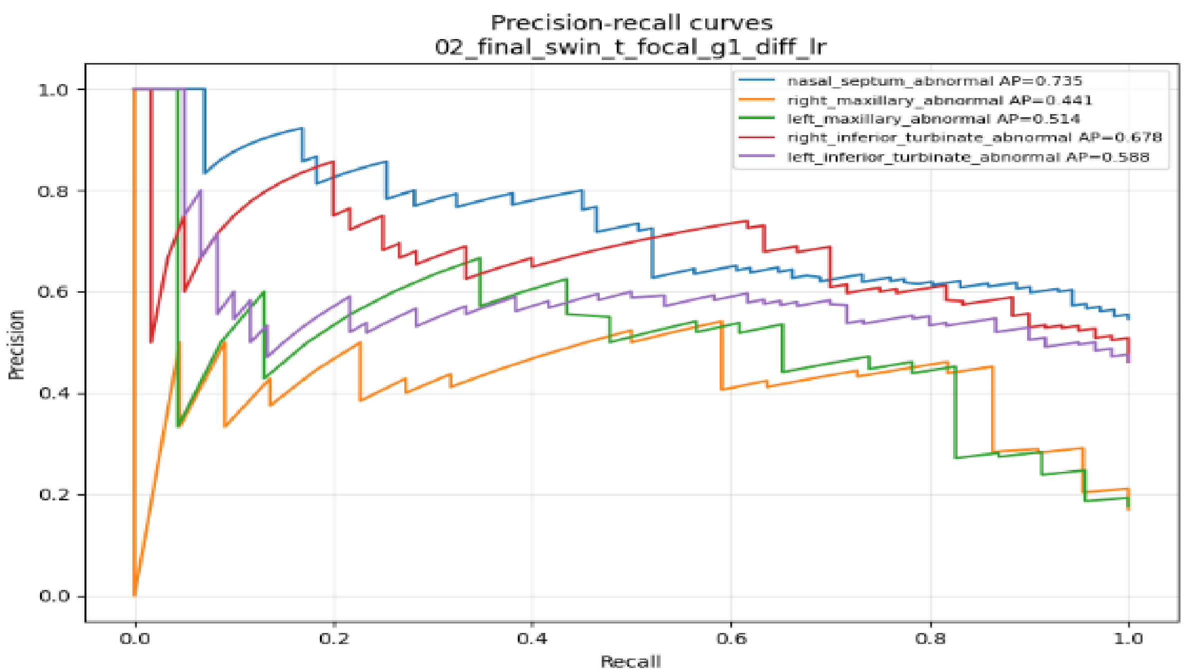

Figure 5.15: Precision-Recall curves for Swin-T

5.5 Comparison with Previous Works

The proposed system differs from many previous works because it combines anatomical segmentation with multi-label abnormality prediction. NasalSeg and similar studies focus mainly on segmentation, while several maxillary-sinus studies focus on a narrower classification target. This project uses segmentation as an intermediate anatomical guide and then predicts multiple abnormality labels. This is a more difficult setting because labels may co-occur and some targets do not have direct segmentation masks.

Table 5.6: Comparison of the proposed model with previous works

Work	Task	Data / Scope	Main Result or Focus	Relation to this project
NasalSeg	Anatomical segmentation	130 CT scans, five structures	Public dataset and segmentation benchmark	Provides the anatomical basis for this project.
Whangbo et al., 2024	Paranasal sinus segmentation	Sinusitis CT data	Multi-class sinus region segmentation	Related segmentation task but not multi-label classification.
Bhattacharya et al., 2024	Anomaly classification	Paranasal/maxillary anomaly focus	Self-supervised representation learning	Related classification direction but narrower label scope.
ConvNeXt	General image classification backbone	Natural image benchmarks	Modern CNN representation learning	Used as final classifier backbone.
Swin Transformer	General transformer vision backbone	Natural image / dense prediction benchmarks	Hierarchical shifted-window attention	Used as final transformer comparison.
Improved ConvNeXt-Tiny medical classification	Medical image classification	General medical image classification	Efficient ConvNeXt-Tiny adaptation	Supports ConvNeXt-Tiny choice but not sinus-specific.
Proposed system	Segmentation-guided multi-label classification	130 CT cases and five abnormality labels	ConvNeXt-Tiny OOF macro F1 = 0.6928	Combines anatomical localization and abnormality prediction.

5.6 Discussion of Results

5.6.1 Effect of Segmentation-Guided ROI Extraction

Segmentation-guided ROI extraction was one of the main reasons the final pipeline became interpretable. Instead of asking the classifier to search the whole CT volume, each label was connected to a specific region. This design is especially important in sinus CT because abnormalities are region-dependent. The segmentation stage therefore acts as an anatomical filter before classification.

5.6.2 Effect of Final ROI Design

The final ROI design avoided fragile artificial masks for structures that NasalSeg does not directly annotate. The septum was modeled using both nasal cavities with padding, while inferior turbinate hypertrophy was modeled using the ipsilateral nasal cavity. These contextual ROIs did not eliminate all difficulty, but they were more stable than pseudo-septum or pseudo-turbinate strategies.

5.6.3 Effect of Imbalance Handling

Class imbalance was handled with focal loss and positive class weighting. These choices helped the model pay more attention to difficult and underrepresented labels. However, imbalance was not fully solved, especially for maxillary abnormalities. This shows that algorithmic weighting can reduce imbalance effects but cannot replace the need for more positive cases.

5.6.4 Why ConvNeXt-Tiny Was Selected

ConvNeXt-Tiny outperformed Swin-T in fold-level macro F1, OOF macro F1, overall label accuracy, and exact match accuracy. The result suggests that a modern convolutional backbone was better suited to the available localized ROI data than the transformer comparison model. Therefore, ConvNeXt-Tiny is selected as the final proposed classifier.

The literature review supports this interpretation. ConvNeXt was designed to modernize CNNs while preserving convolutional inductive bias, and several medical-imaging works have adapted ConvNeXt-style designs to limited-data medical settings. Therefore, the choice of ConvNeXt-Tiny is supported both experimentally in this project and conceptually by recent medical-imaging research.

5.6.5 Maxillary Label Instability

The right and left maxillary labels were the most unstable labels. The main reason is the low number of positive cases: 22 for right maxillary abnormality and 23 for left maxillary abnormality. These labels also frequently co-occur with septal or turbinate abnormalities. Co-occurrence makes it harder to isolate the contribution of each label and can make the dominant-label confusion matrix appear worse than the full multi-label prediction actually is.

5.6.6 Limitations of the 5×5 Dominant-Label Confusion Matrix

The 5×5 matrix is useful as a compact visualization, but it is not the main selection metric. It compresses a multi-label case into one dominant true label and one dominant predicted label. This can hide partially correct predictions. For example, a case with both maxillary and septal abnormalities may be counted as confused if the model predicts the correct maxillary label but gives higher confidence to the co-occurring septal label. Therefore, OOF macro F1, per-label F1, and threshold-tuned metrics are more appropriate for evaluating the final model.

5.7 Chapter Summary

The results show that SwinUNETR provides strong anatomical segmentation and that ConvNeXt-Tiny is the best final classifier among the evaluated models. The main remaining weaknesses are related to label imbalance, multi-label co-occurrence, and lack of direct masks for septum and turbinates. These issues guide the future work described in the final chapter.

Chapter Six: Conclusion

6.1 Introduction

This chapter summarizes the main findings of the project, states the conclusions, identifies the research limitations, and proposes future directions. The project should be viewed as a research prototype for segmentation-guided sinus CT analysis rather than a clinically deployed system.

6.2 Conclusions

This project developed a segmentation-assisted AI system for sinus CT analysis. The system uses SwinUNETR to segment key anatomical structures, extracts anatomy-guided regions of interest, and applies multi-label classification to predict five selected abnormalities. ConvNeXt-Tiny was selected as the final classifier because it achieved the best overall performance among the final compared models.

The segmentation model achieved strong performance, with a mean Dice score of 0.8743 and mean IoU of 0.8102 on the test set. These results indicate that the model can localize the available NasalSeg structures well enough to support downstream ROI extraction. The classification model achieved a ConvNeXt-Tiny OOF macro F1 score of 0.6928, outperforming Swin-T on the final evaluation. The results demonstrate that the proposed approach is promising but also data-limited.

The study also shows that evaluation must be interpreted carefully in multi-label medical imaging tasks. The 5×5 dominant-label confusion matrix is visually helpful but underrepresents multi-label correctness, especially for cases with co-occurring maxillary, septal, and turbinate abnormalities. Therefore, per-label and out-of-fold metrics are more appropriate for final model selection.

6.3 Research Limitations

- The dataset contains only 130 matched CT cases, which limits the generalization of deep learning models.
- The right and left maxillary abnormality labels have low positive counts compared with other labels.
- NasalSeg does not directly segment nasal septum or turbinate structures, requiring contextual ROI approximations.
- The classification task is multi-label, and co-occurrence makes dominant-label visualization incomplete.
- The final model was not externally validated on data from another hospital or scanner.

• The system is a research prototype and should not be considered clinically deployable without further validation.

6.4 Challenges and Future Work

• Increase the dataset size, especially for right and left maxillary abnormality labels.

• Add direct expert annotations for the nasal septum and turbinates to reduce dependence on contextual proxy ROIs.

• Validate the final model on an external dataset from another scanner or hospital.

• Explore 3D or hybrid 2.5D/3D classifiers after increasing the number of labeled cases.

• Improve label quality through radiologist review and standardized abnormality definitions.

• Evaluate calibration and uncertainty estimation to make predictions more clinically interpretable.

• Develop a user interface that displays CT slices, segmentation masks, ROIs, and predicted abnormalities together.

• Use multi-label evaluation dashboards rather than relying only on dominant-label confusion matrices.

Table 6.1: Summary of main findings and future work directions

Finding	Implication	Future direction
SwinUNETR segmentation performed well	Segmentation can guide ROI extraction	Use improved/manual masks for more structures.
ConvNeXt-Tiny outperformed Swin-T	Modern CNNs were more stable for this dataset	Keep ConvNeXt-Tiny as final baseline.
Maxillary labels were unstable	Low positive counts and co-occurrence affect learning	Collect more maxillary-positive cases.
Dominant-label matrix is limited	Multi-label correctness may be hidden	Use per-label and case-level multi-label metrics.
Septum/turbinate masks were unavailable	Contextual ROIs are approximations	Annotate septum and turbinate structures directly.

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